

Code

Jayden Zhou

Predict the average reaction time through the variables we get from the survey

```
library(leaps)
```

```
## Warning: package 'leaps' was built under R version 4.3.3
```

```
library(olsrr)
```

```
## Warning: package 'olsrr' was built under R version 4.3.3
```

```
##
```

```
## Attaching package: 'olsrr'
```

```
## The following object is masked from 'package:datasets':
```

```
##
```

```
## rivers
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.1.4      v readr      2.1.5
```

```
## v forcats   1.0.0      v stringr   1.5.1
```

```
## v ggplot2    3.5.1      v tibble    3.2.1
```

```
## v lubridate  1.9.3      v tidyr     1.3.1
```

```
## v purrr      1.0.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggplot2)
```

```
library(glmnet)
```

```
## Warning: package 'glmnet' was built under R version 4.3.3
```

```
## Loading required package: Matrix
```

```
##
```

```
## Attaching package: 'Matrix'
```

```
##
```

```
## The following objects are masked from 'package:tidyr':  
##  
##   expand, pack, unpack  
##  
## Loaded glmnet 4.1-8
```

```
library(lmtest)
```

```
## Warning: package 'lmtest' was built under R version 4.3.3
```

```
## Loading required package: zoo
```

```
## Warning: package 'zoo' was built under R version 4.3.3
```

```
##  
## Attaching package: 'zoo'  
##  
## The following objects are masked from 'package:base':  
##  
##   as.Date, as.Date.numeric
```

```
library(nlme)
```

```
## Warning: package 'nlme' was built under R version 4.3.3
```

```
##  
## Attaching package: 'nlme'  
##  
## The following object is masked from 'package:dplyr':  
##  
##   collapse
```

```
library(car)
```

```
## Warning: package 'car' was built under R version 4.3.3
```

```
## Loading required package: carData
```

```
## Warning: package 'carData' was built under R version 4.3.3
```

```
##  
## Attaching package: 'car'  
##  
## The following object is masked from 'package:dplyr':  
##  
##   recode  
##  
## The following object is masked from 'package:purrr':  
##  
##   some
```

```
set.seed(123)
```

```
# Read in the data
Survey <- read.csv("Survey.csv")
names(Survey)
```

```
## [1] "First"          "Second"          "Third"           "Class"
## [5] "Age"            "AvgSleepTime"    "LastNightSleep"  "HoursAweak"
## [9] "Fatigue"        "Stress"          "Distraction"     "NoiseLevel"
## [13] "Temperature"    "VideoGamePlay"   "SportsPlay"      "ShootingGames"
## [17] "AvgHoursExercise" "Gamer"           "CaffeinIntake"   "AlcoholIntake"
## [21] "VisualAcuity"   "PrimarilyHand"   "UsePrimaryHand"  "Cautious"
## [25] "InputDevice"    "RefreshRate"     "DeviceOS"        "WiFi"
```

```
# Data Cleaning: IQR Outliers
```

```
## Define function
```

```
remove_outliers_iqr <- function(df, columns) {
  for (col in columns) {
    Q1 <- quantile(df[[col]], 0.25, na.rm = TRUE)
    Q3 <- quantile(df[[col]], 0.75, na.rm = TRUE)
    IQR_value <- Q3 - Q1
    lower_bound <- Q1 - 1.5 * IQR_value
    upper_bound <- Q3 + 1.5 * IQR_value
    df <- df[df[[col]] >= lower_bound & df[[col]] <= upper_bound, ]
  }
  return(df)
}
```

```
# Create response Y -- mean reaction time of three tests
```

```
survey <- Survey
survey$avg_time <- rowMeans(Survey[, c("First", "Second", "Third")])
```

```
## New data frame with no outliers
```

```
survey_cleaned <- remove_outliers_iqr(survey, c("avg_time", "AvgSleepTime", "LastNightSleep", "HoursAweak"))
```

```
## Remove not informative column
```

```
survey_cleaned <- survey_cleaned %>%
  dplyr::select(-AlcoholIntake)
```

```
# Factorize --> levels
```

```
survey_cleaned <- survey_cleaned %>%
  mutate(
    Fatigue = factor(Fatigue, levels = c("Extremely fatigued", "Very fatigued", "Moderately fatigued", "Slightly fatigued", "Not fatigued")),
    Stress = factor(Stress, levels = c("Very High", "High", "Moderate", "Low", "Very Low")),
    Cautious = factor(Cautious, levels = c("Extremely cautious", "Very cautious", "Moderately cautious", "Slightly cautious", "Not cautious")),
    WiFi = factor(WiFi, levels = c("Unstable", "Stable")),
    RefreshRate = factor(RefreshRate, levels = c("High", "Mid-Range", "Standard")),
    InputDevice = factor(InputDevice, levels = c("Game controller", "Mouse", "Keyboard", "Trackpad", "Touchscreen")),
    UsePrimaryHand = factor(UsePrimaryHand, levels = c("Yes", "No")),
    PrimarilyHand = factor(PrimarilyHand, levels = c("Right hand", "Left hand", "Ambidextrous (both hands)")),
    VideoGamePlay = factor(VideoGamePlay, levels = c("Daily", "Several times a week", "Once a week", "Several times a month", "Once a month", "Never")),
    SportsPlay = factor(SportsPlay, levels = c("Daily", "Several times a week", "Once a week", "Several times a month", "Once a month", "Never"))
```

```

ShootingGames = factor(ShootingGames, levels = c("None", "Beginner", "Intermediate", "Advanced", "Expert"))
Temperature = factor(Temperature, levels = c("Very Warm", "Warm", "Neutral", "Cold"))
Class = factor(Class, levels = c("Sophomore", "Junior", "Senior", "Graduate student"))
)

```

```

# Exploratory Data Analysis
## Continuous variables summary
summary(survey_cleaned)

```

```

##           First           Second           Third           Class
## Min.      :325.0    Min.      :338.0    Min.      :341.0    Sophomore      :45
## 1st Qu.:520.0    1st Qu.:498.0    1st Qu.:490.0    Junior         :55
## Median :593.0    Median :561.0    Median :542.0    Senior         :22
## Mean   :592.5    Mean   :560.7    Mean   :547.7    Graduate student: 7
## 3rd Qu.:663.0    3rd Qu.:633.0    3rd Qu.:613.0
## Max.    :934.0    Max.    :833.0    Max.    :783.0
##           Age           AvgSleepTime           LastNightSleep           HoursAweek
## Min.      :18.0    Min.      : 5.000    Min.      : 4.000    Min.      : 0.000
## 1st Qu.:20.0    1st Qu.: 6.500    1st Qu.: 6.000    1st Qu.: 4.000
## Median :20.0    Median : 7.000    Median : 7.000    Median : 6.000
## Mean   :20.9    Mean   : 7.236    Mean   : 7.264    Mean   : 7.322
## 3rd Qu.:21.0    3rd Qu.: 8.000    3rd Qu.: 8.000    3rd Qu.:10.000
## Max.    :26.0    Max.    :10.000    Max.    :11.000    Max.    :18.000
##           Fatigue           Stress           Distraction           NoiseLevel
## Extremely fatigued : 1    Very High: 5    Length:129    Min.      : 1.000
## Very fatigued      :16    High       :20    Class :character 1st Qu.: 2.000
## Moderately fatigued:44    Moderate :46    Mode  :character Median : 3.000
## Slightly Fatigued  :45    Low       :35    Mean   : 3.287
## Not fatigued at all:23    Very Low :23    3rd Qu.: 5.000
## Max.    :10.000
##           Temperature           VideoGamePlay           SportsPlay
## Very Warm: 2    Daily           :11    Daily           : 5
## Warm       :16    Several times a week :36    Several times a week :24
## Neutral    :98    Once a week           : 9    Once a week           :11
## Cold       :13    Several times a month:24    Several times a month:24
## Rarely           :38    Rarely           :50
## Never           :11    Never           :15
##           ShootingGames AvgHoursExercise           Gamer           CaffeinIntake
## None          :19    Min.      : 0.000    Length:129    Length:129
## Beginner      :54    1st Qu.: 2.000    Class :character Class :character
## Intermediate:35    Median : 4.000    Mode  :character Mode  :character
## Advanced      :18    Mean   : 4.097
## Expert        : 3    3rd Qu.: 6.000
## Max.          :12.000
##           VisualAcuity           PrimarilyHand UsePrimaryHand
## Length:129    Right hand           :119    Yes:122
## Class :character Left hand           : 6    No : 7
## Mode  :character Ambidextrous (both hands equally): 4
##
##
##
##           Cautious           InputDevice           RefreshRate
## Extremely cautious : 3    Game controller: 1    High           :29

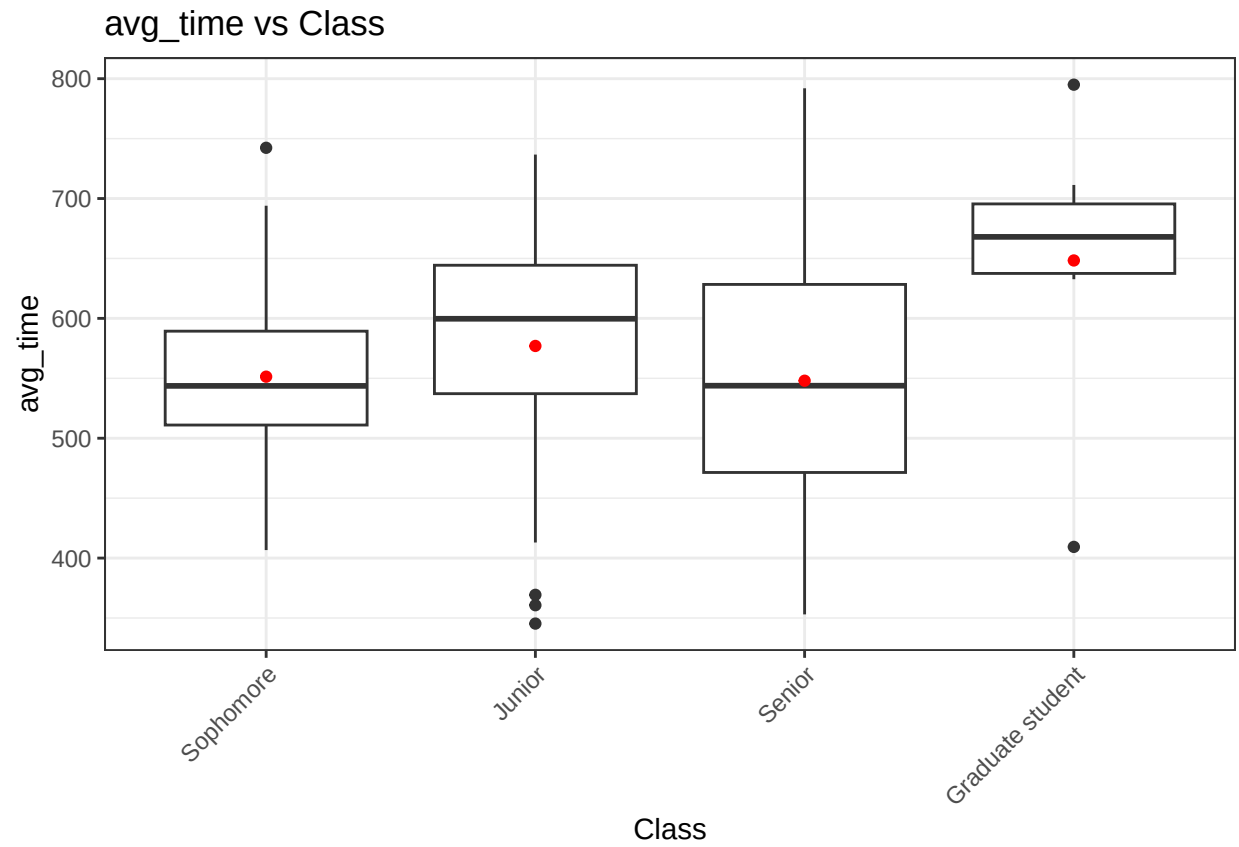
```

```
## Very cautious      :32  Mouse      :47  Mid-Range:52
## Moderately cautious:68  Keyboard   : 2  Standard :48
## Slightly cautious  :21  Trackpad    :69
## Not cautious at all: 5   Touch screen :10
##
## DeviceOS           WiFi           avg_time
## Length:129         Unstable: 2    Min.      :345.3
## Class :character   Stable :127    1st Qu.:511.0
## Mode :character                    Median :574.0
##                                       Mean   :567.0
##                                       3rd Qu.:635.0
##                                       Max.   :795.0
```

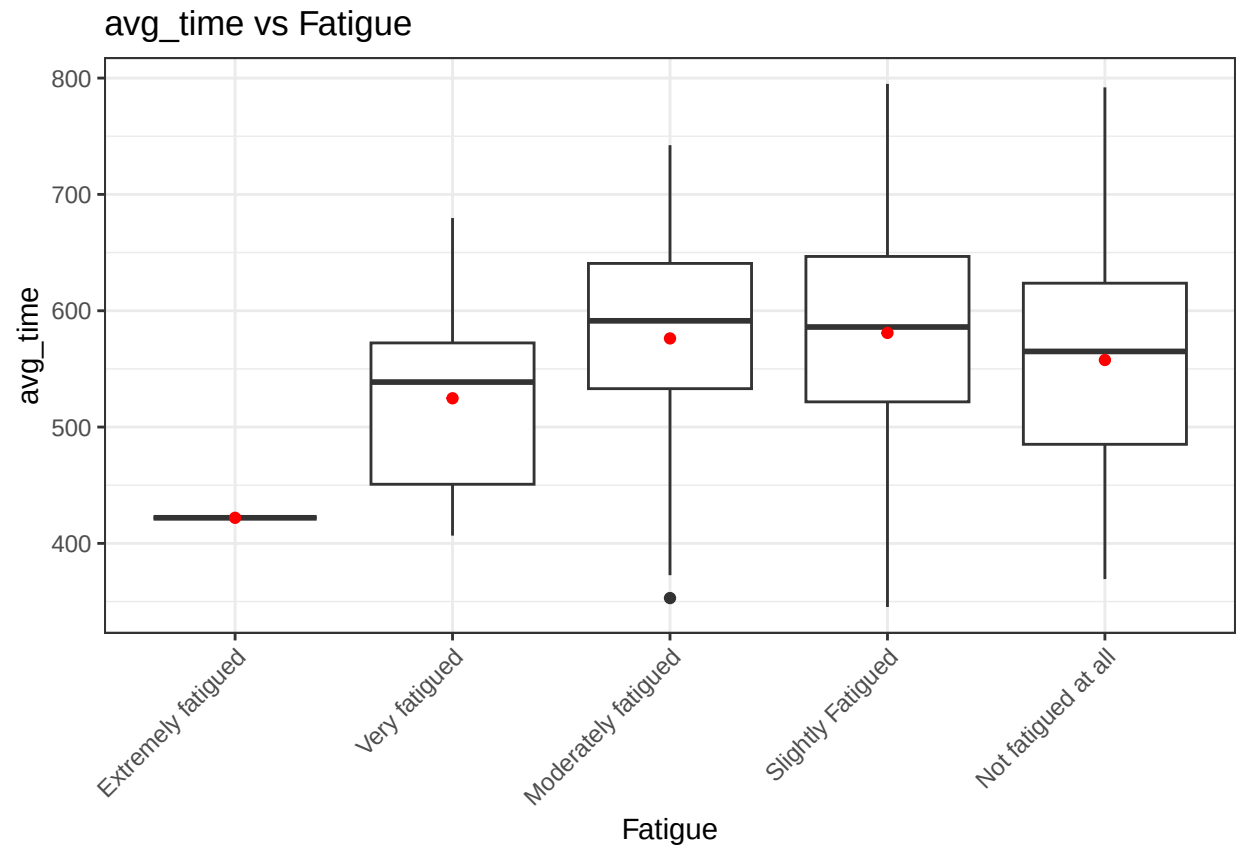
```
## Boxplot function
plot_box <- function(var) {
  survey_cleaned %>%
    ggplot(aes_string(x = var, y = "avg_time")) +
    geom_boxplot() +
    stat_summary(geom = "point", fun = "mean", color = "red") +
    labs(title = paste("avg_time vs", var)) +
    theme_bw() +
    theme(axis.text.x = element_text(angle = 45, hjust = 1))
}

## avg_time vs Class
plot_box("Class")
```

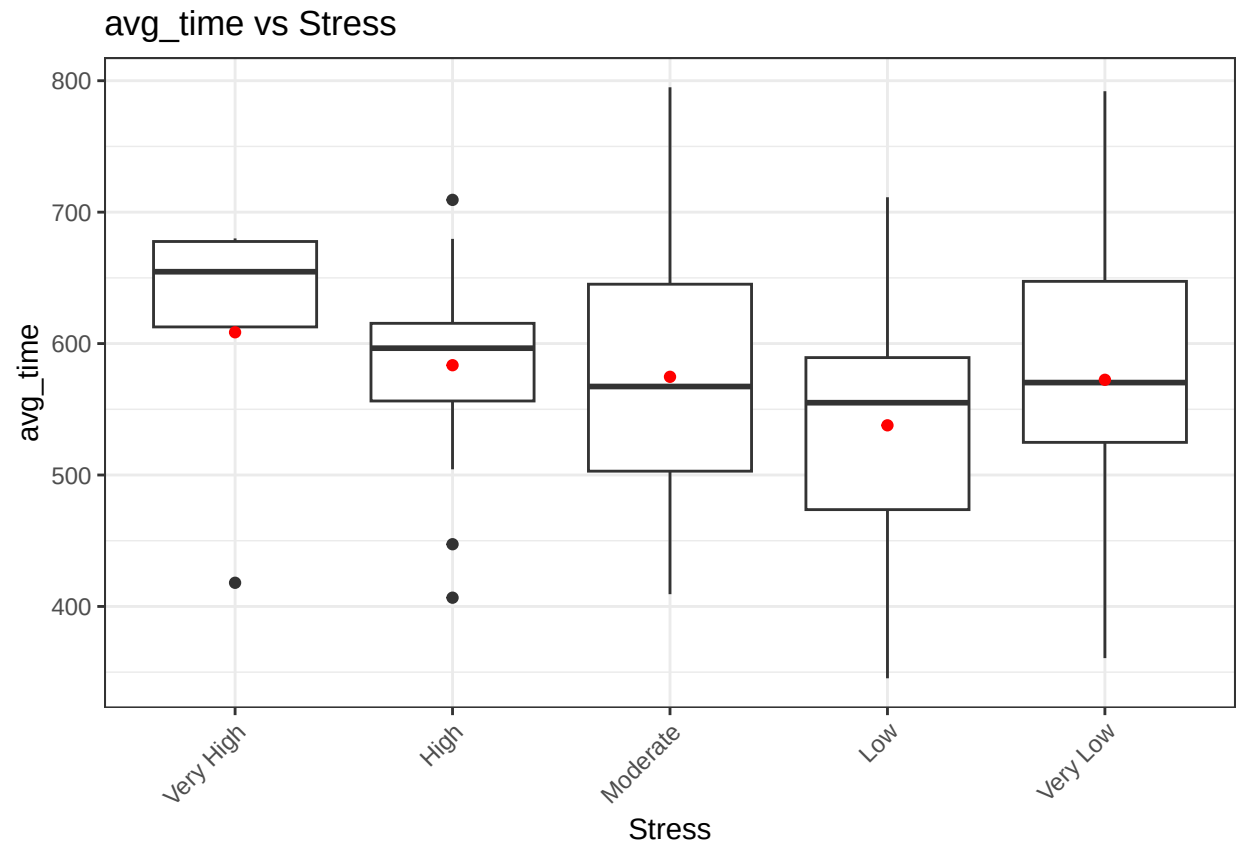
```
## Warning: 'aes_string()' was deprecated in ggplot2 3.0.0.
## i Please use tidy evaluation idioms with 'aes()'.
## i See also 'vignette("ggplot2-in-packages")' for more information.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```



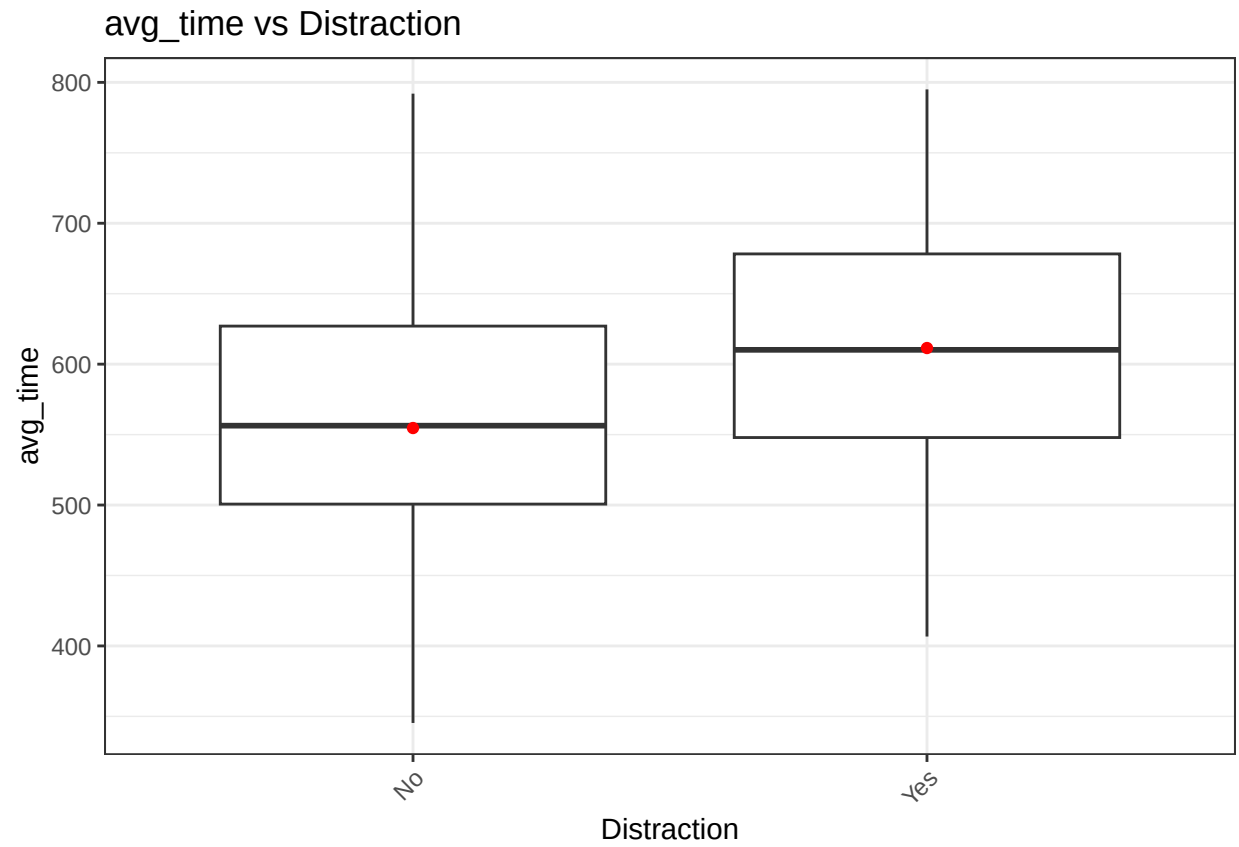
```
## avg_time vs Fatigue  
plot_box("Fatigue")
```



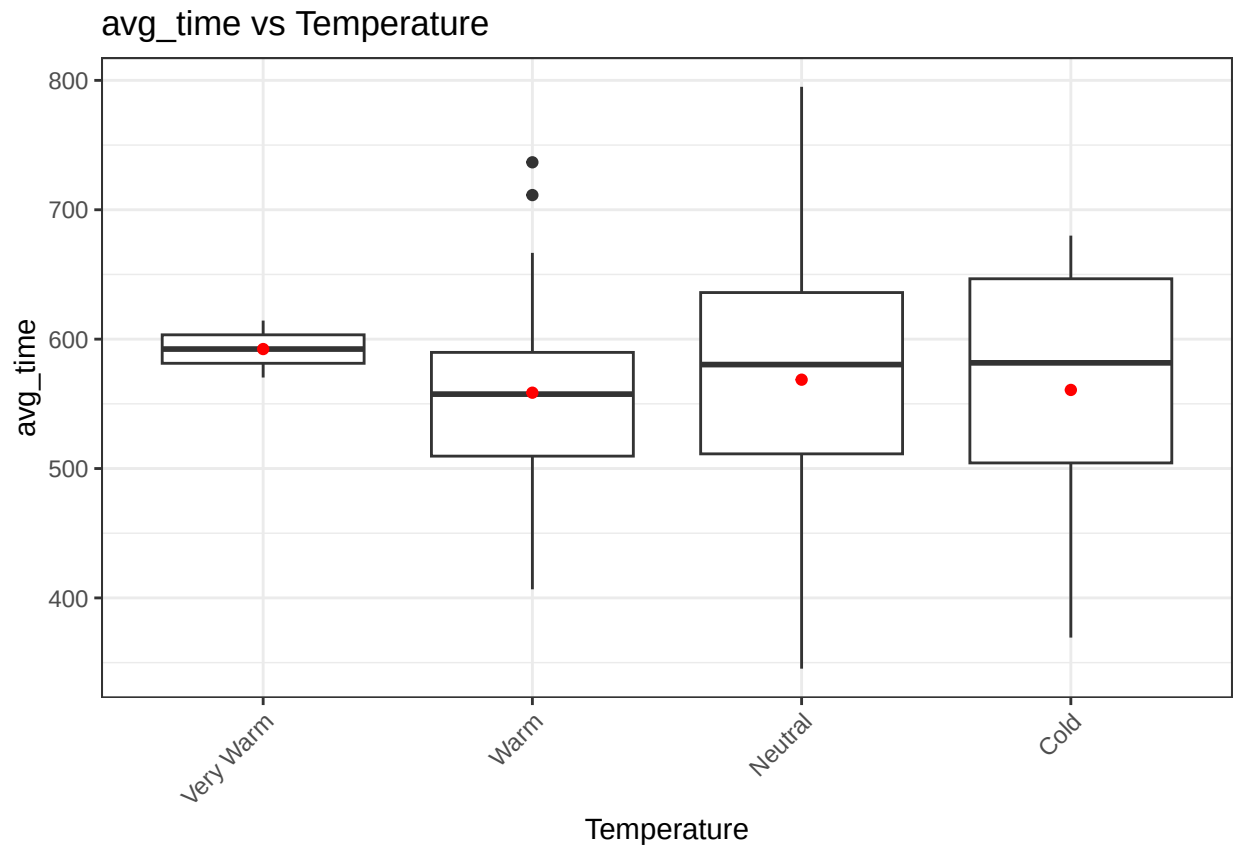
```
## avg_time vs Stress  
plot_box("Stress")
```



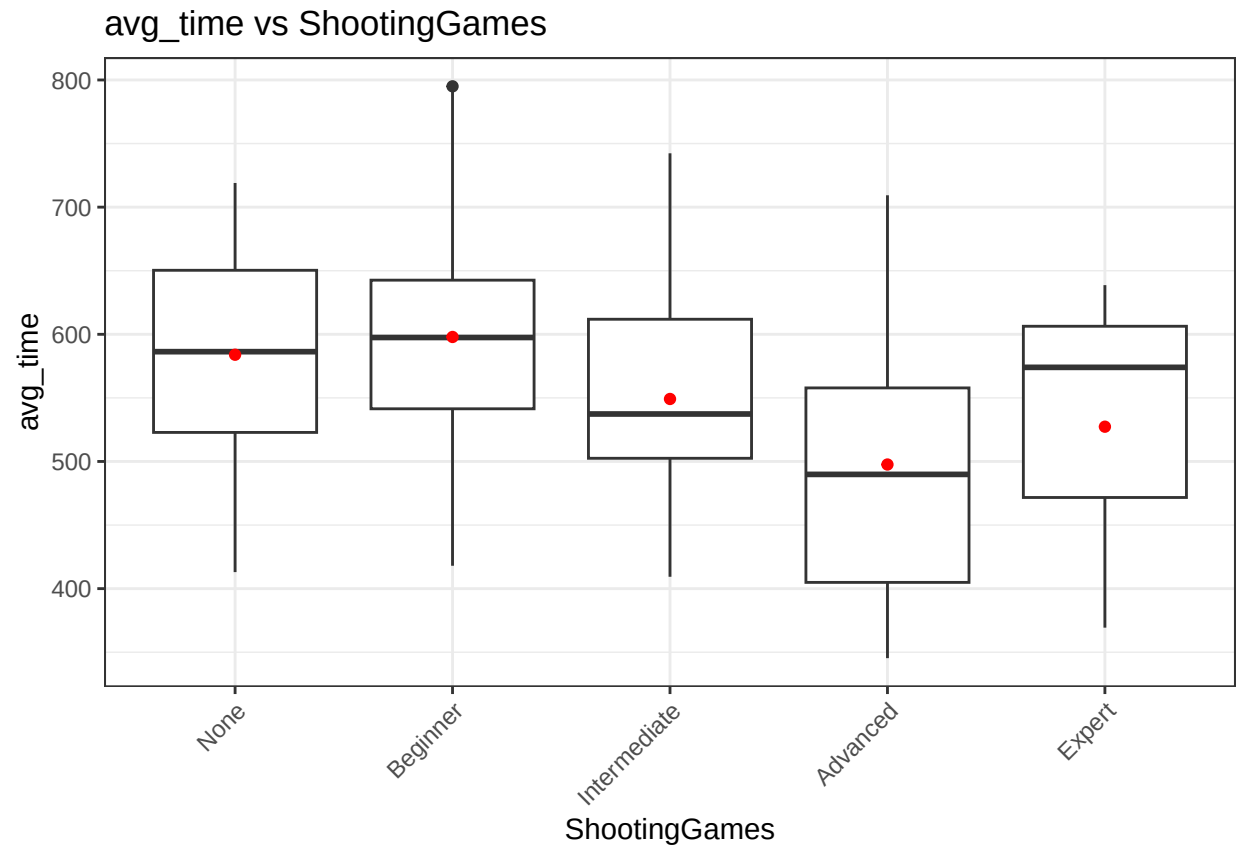
```
## avg_time vs Distraction  
plot_box("Distraction")
```

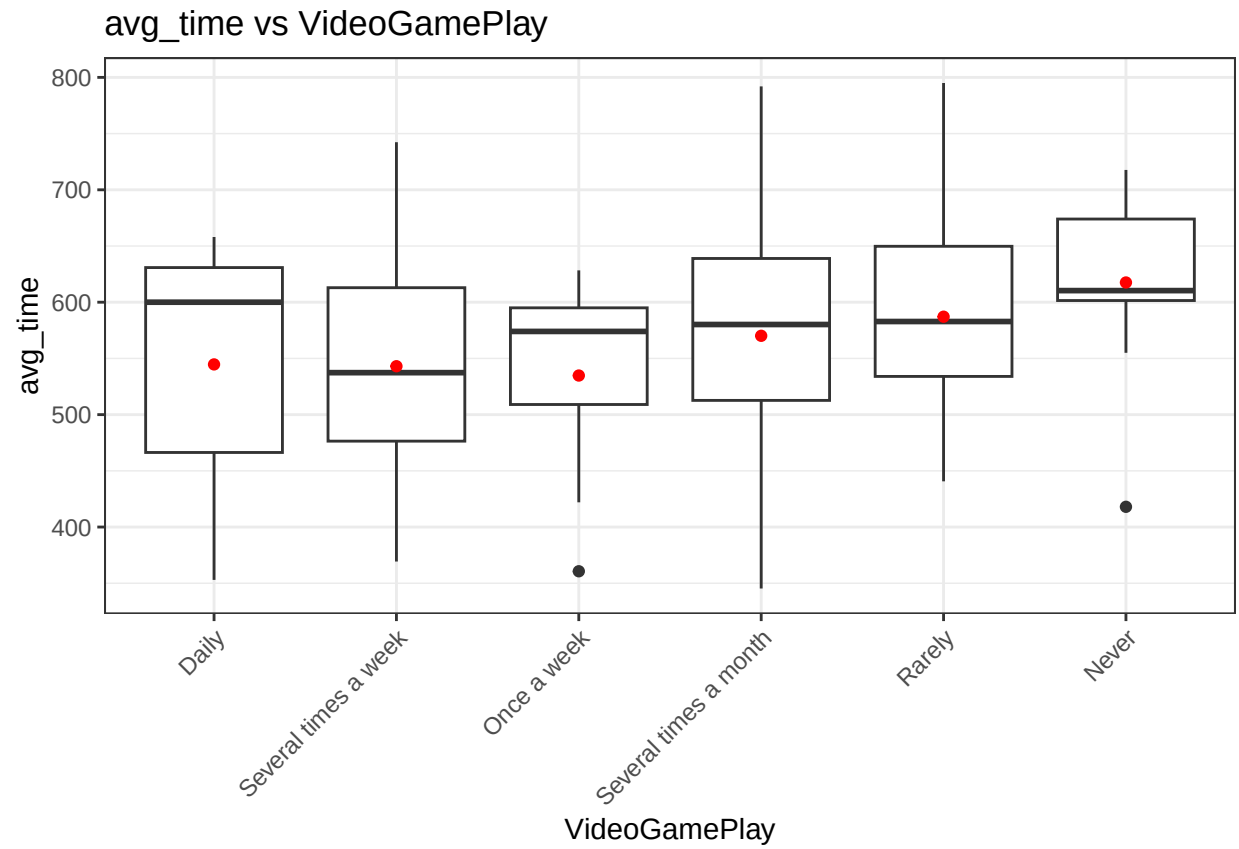
```
## avg_time vs Temperature  
plot_box("Temperature")
```



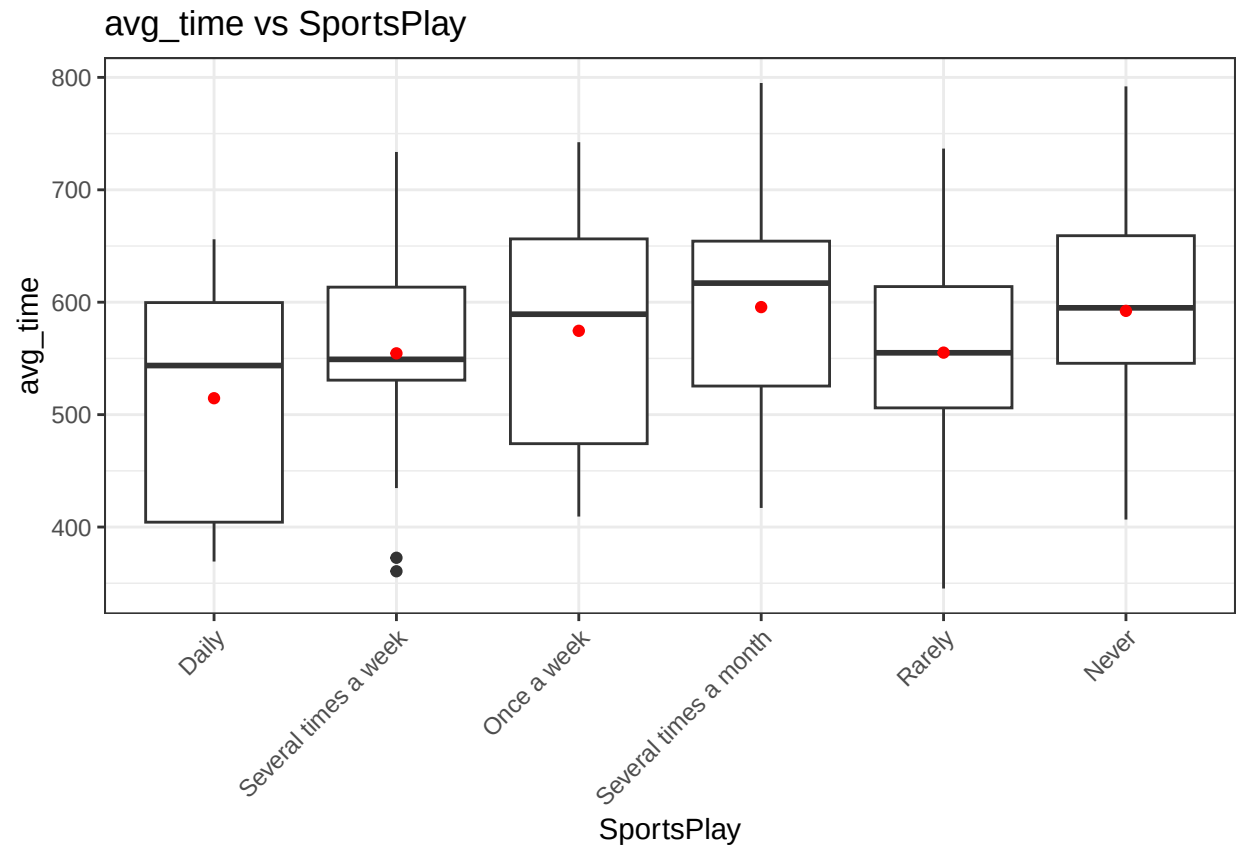
```
## avg_time vs ShootingGames  
plot_box("ShootingGames")
```



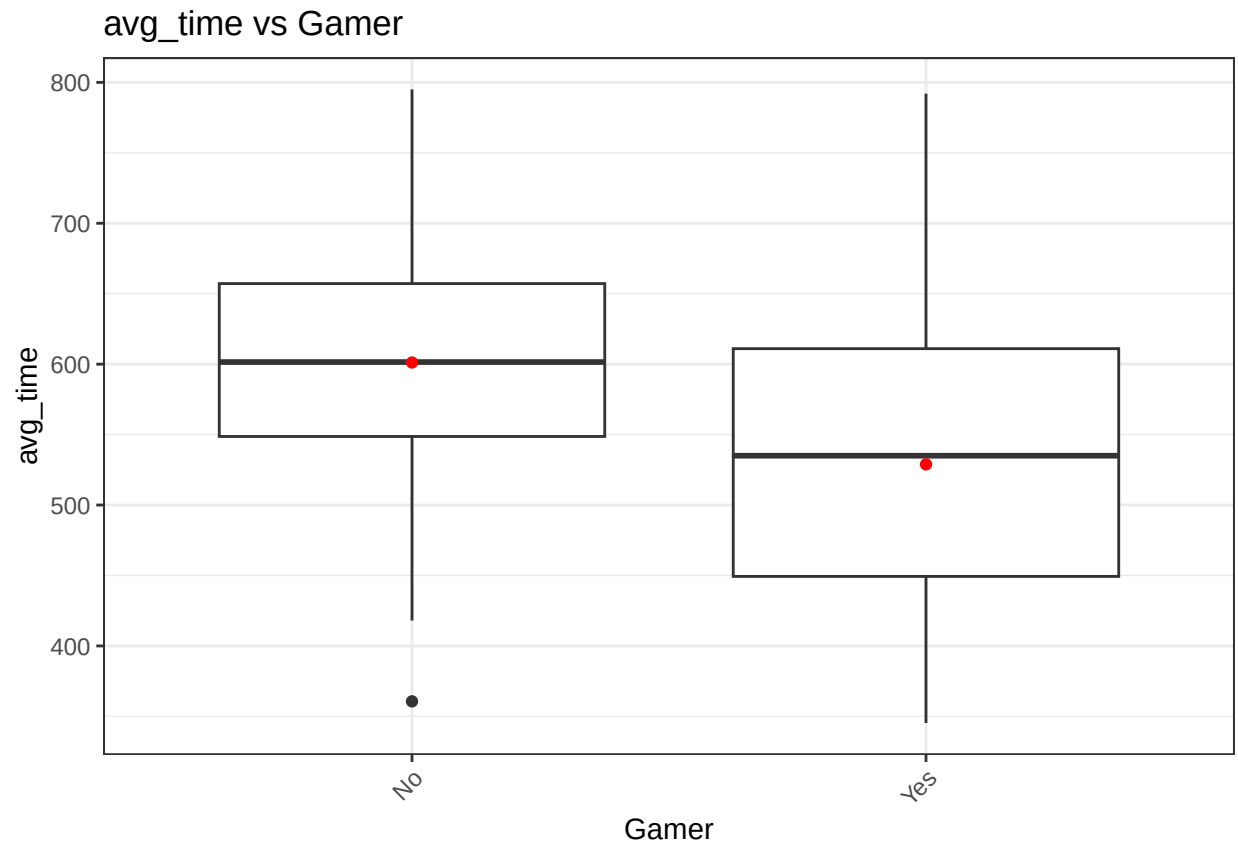
```
## avg_time vs VideoGamePlay  
plot_box("VideoGamePlay")
```



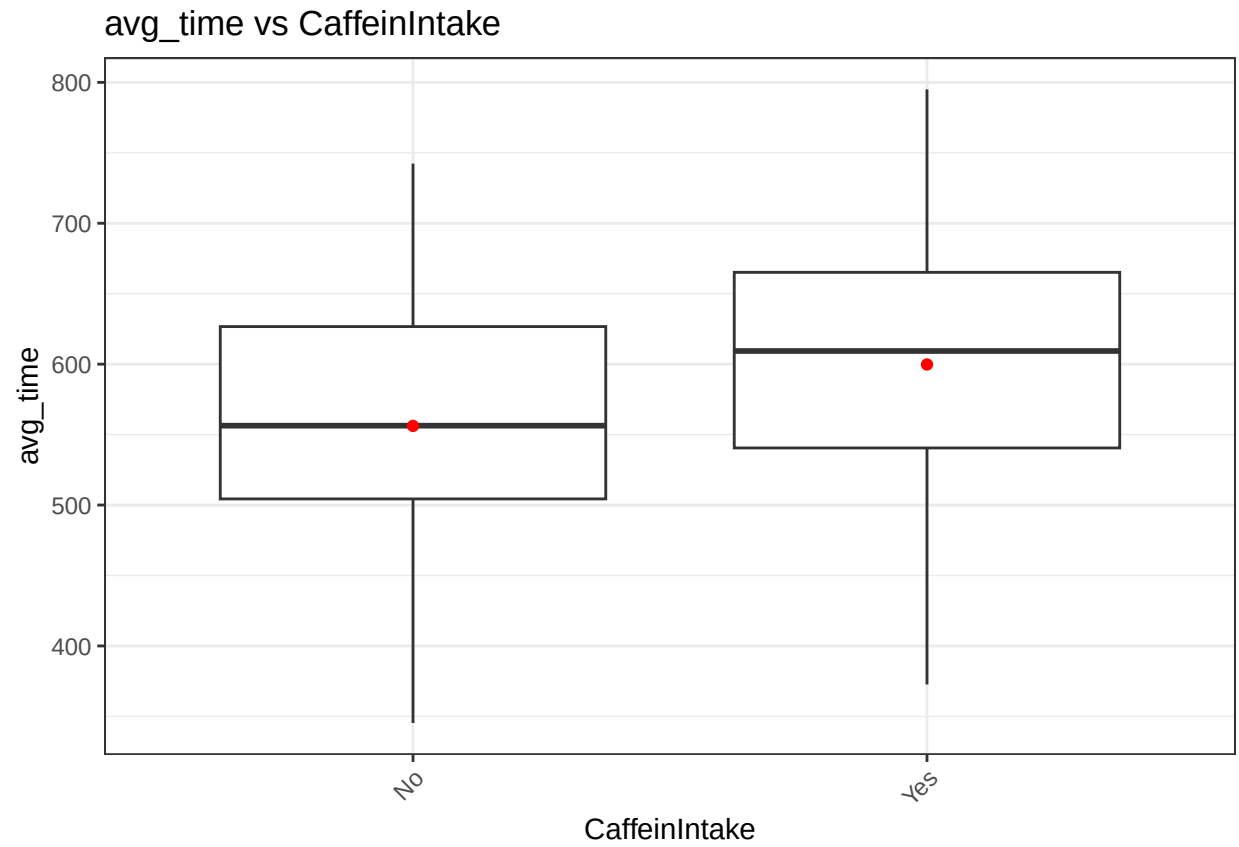
```
## avg_time vs SportsPlay  
plot_box("SportsPlay")
```



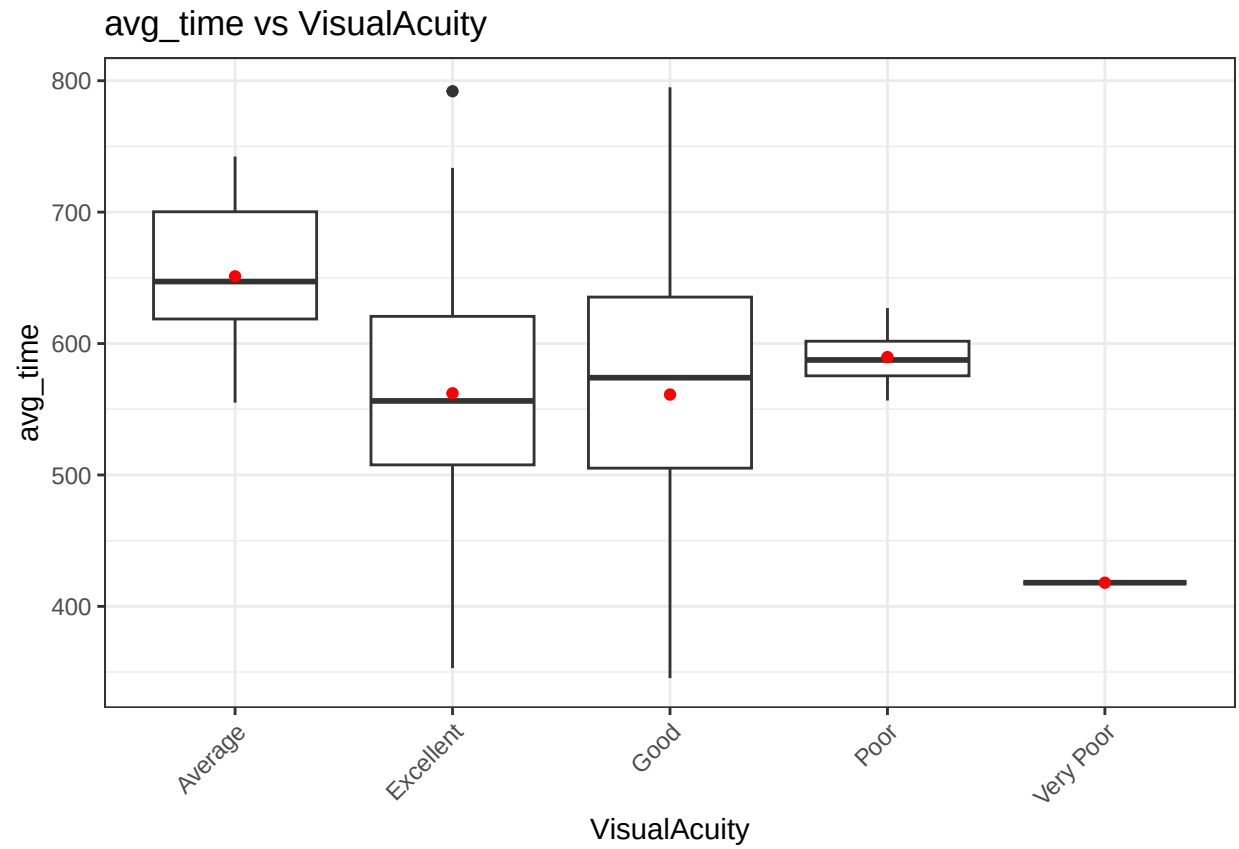
```
## avg_time vs Gamer  
plot_box("Gamer")
```



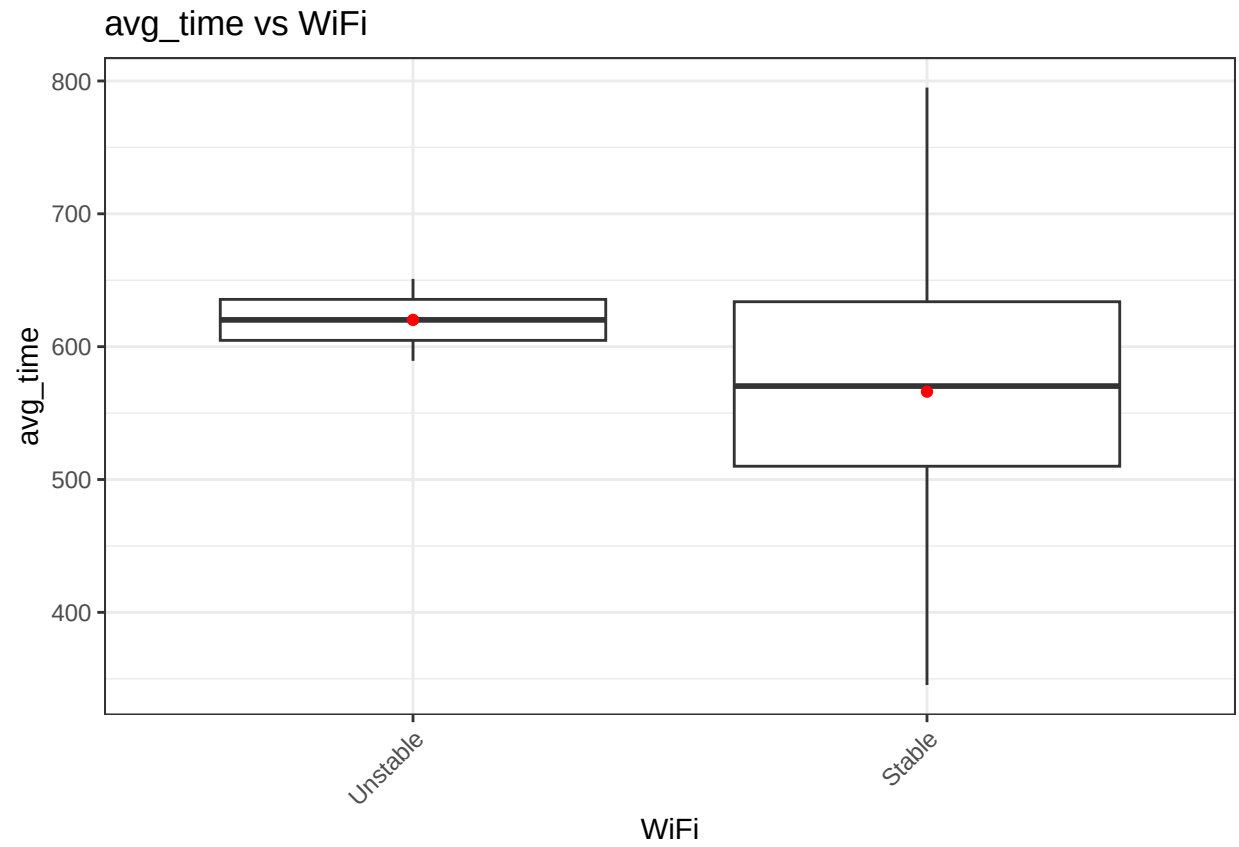
```
## avg_time vs CaffeinIntake  
plot_box("CaffeinIntake")
```



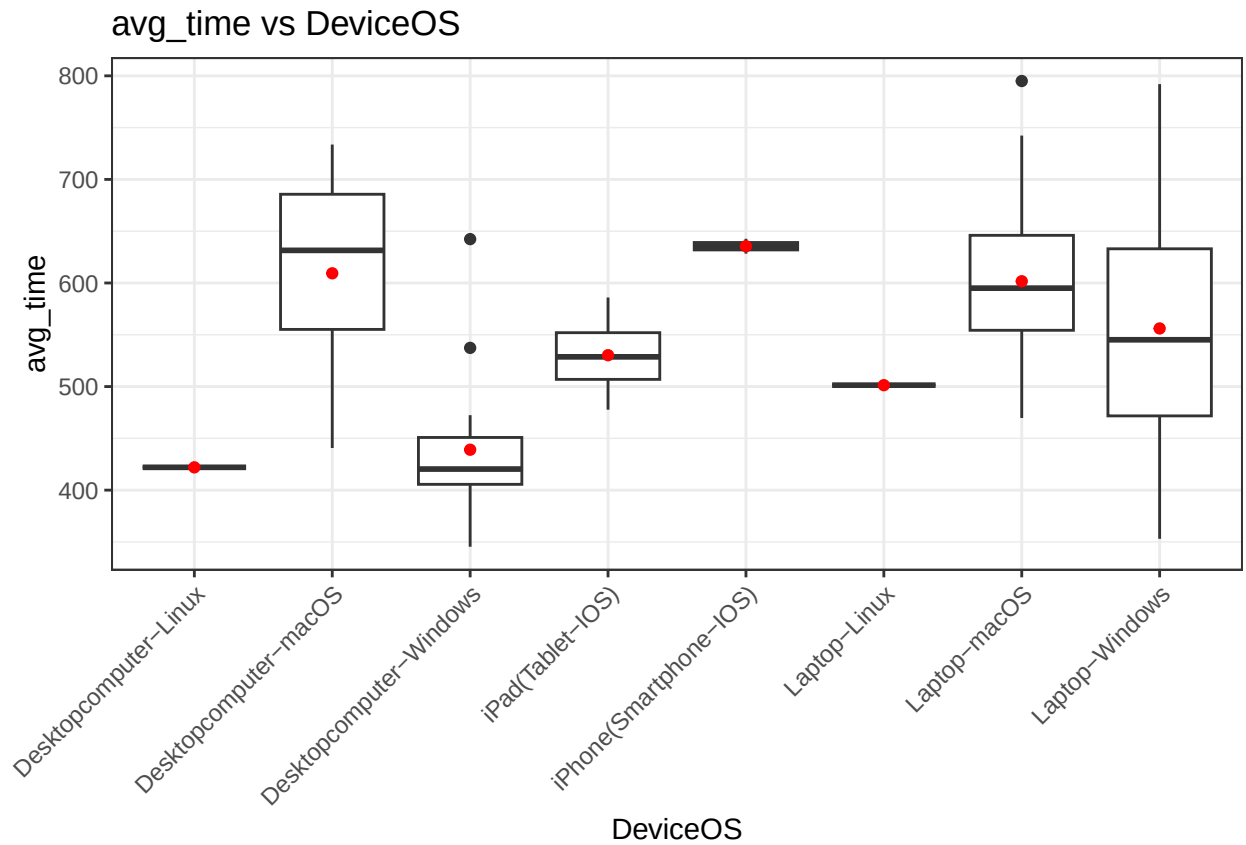
```
## avg_time vs VisualAcuity  
plot_box("VisualAcuity")
```



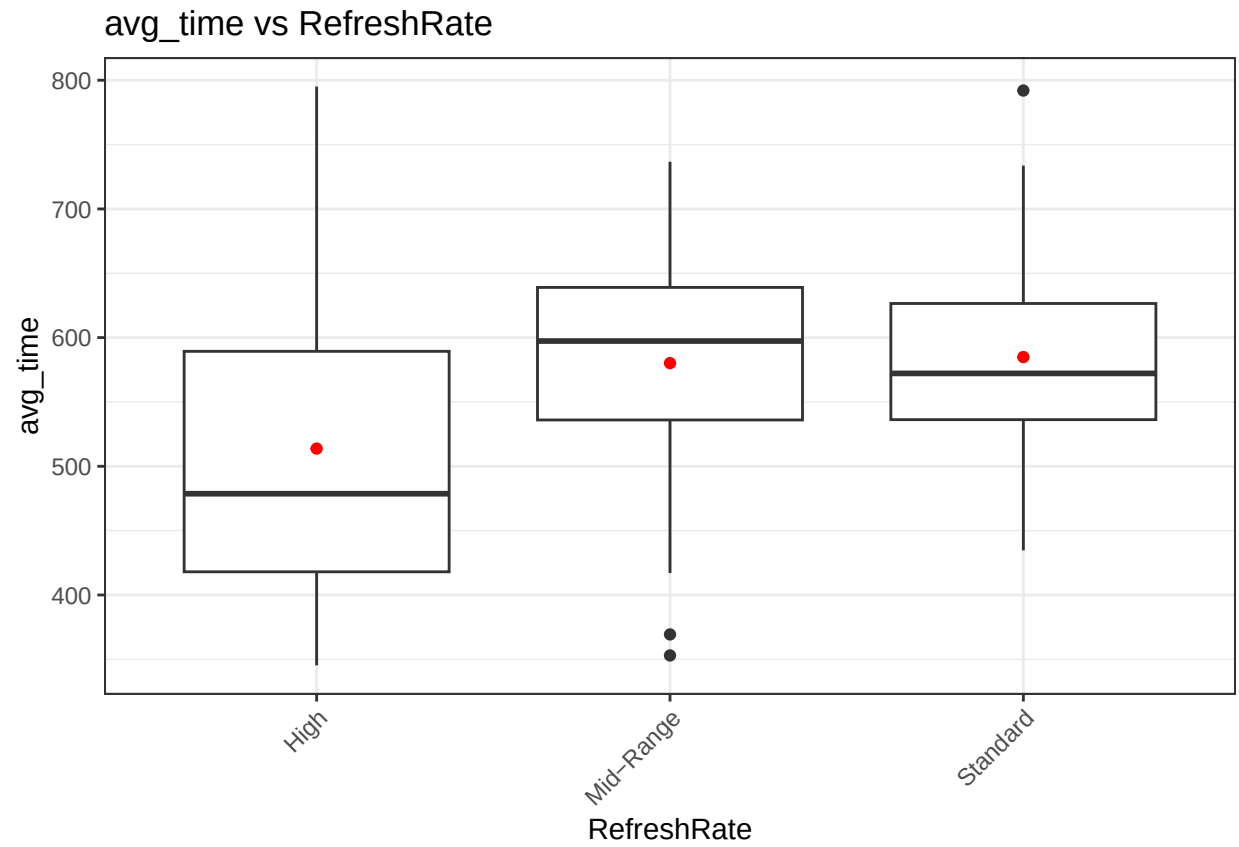
```
## avg_time vs WiFi  
plot_box("WiFi")
```

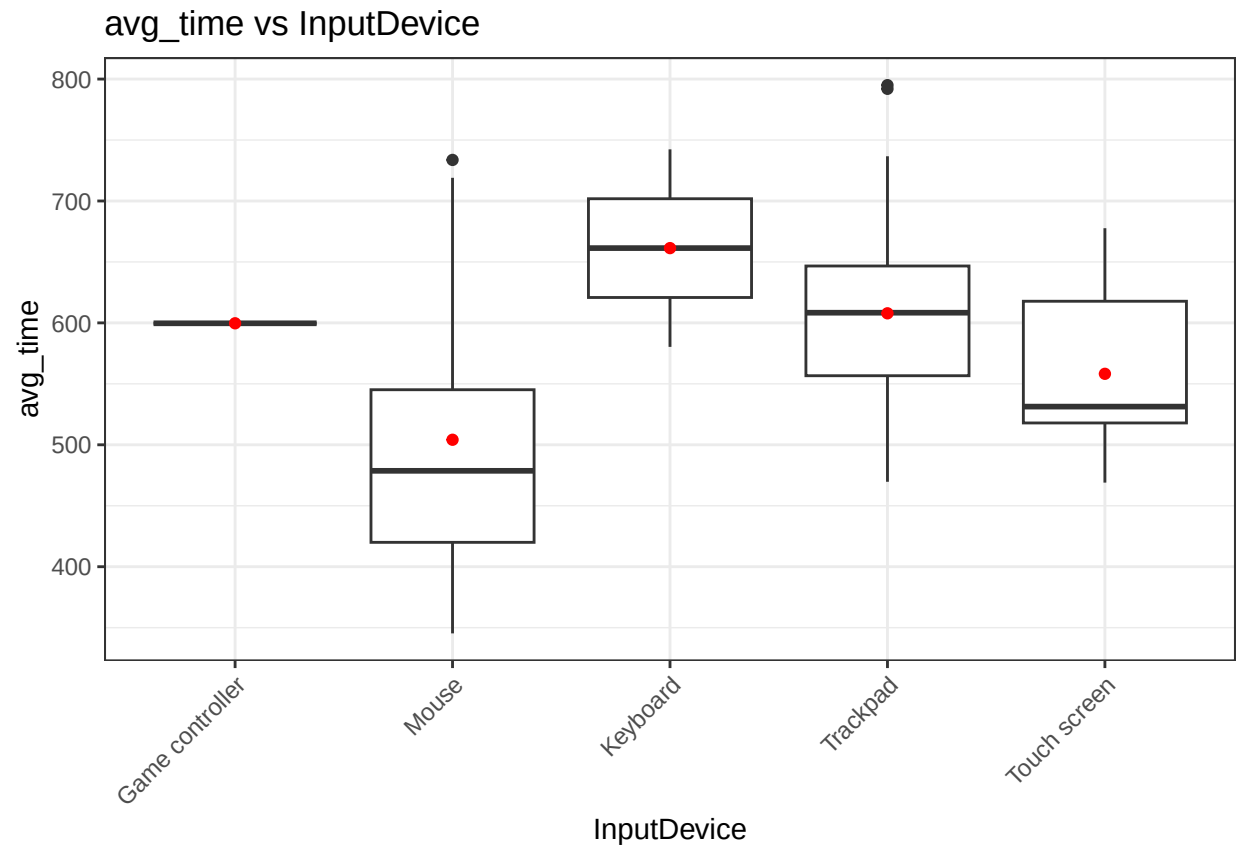
```
## avg_time vs DeviceOS  
plot_box("DeviceOS")
```



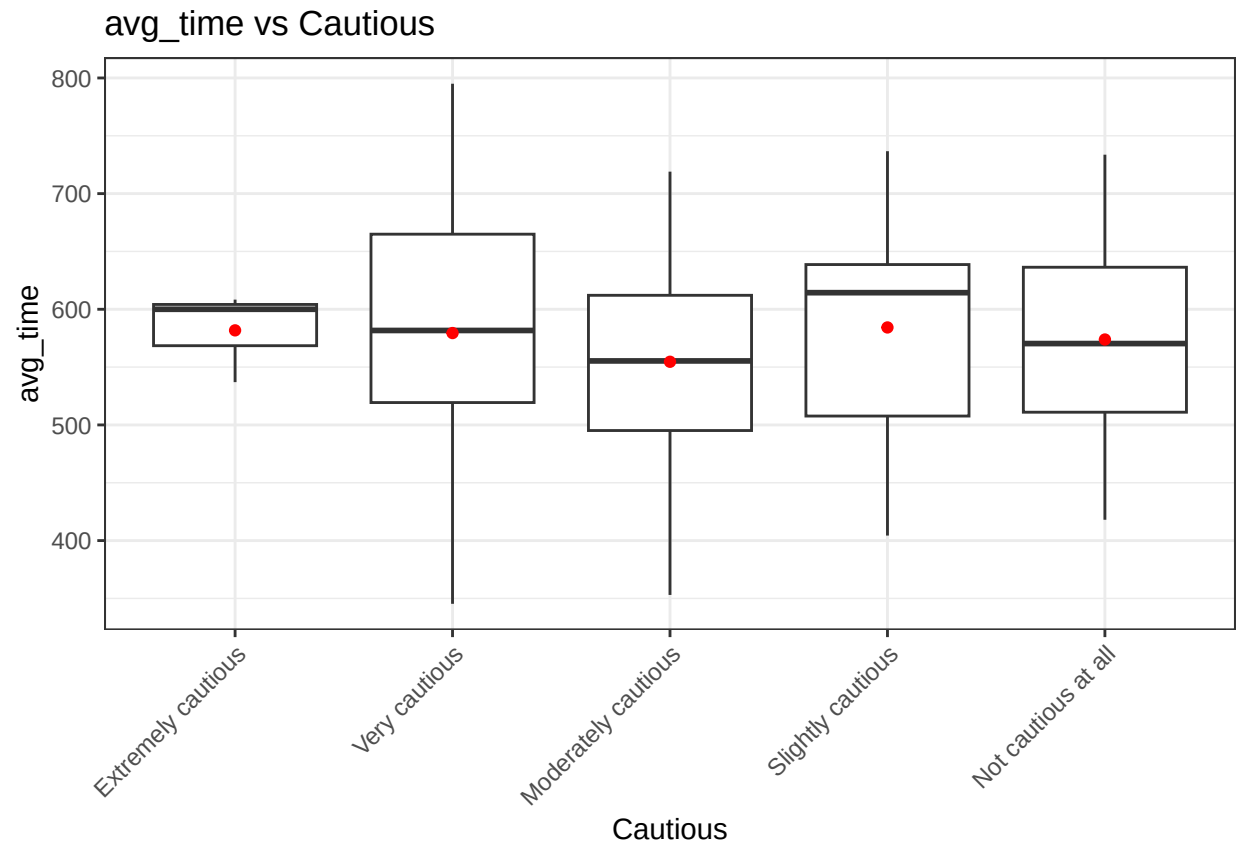
```
## avg_time vs RefreshRate  
plot_box("RefreshRate")
```



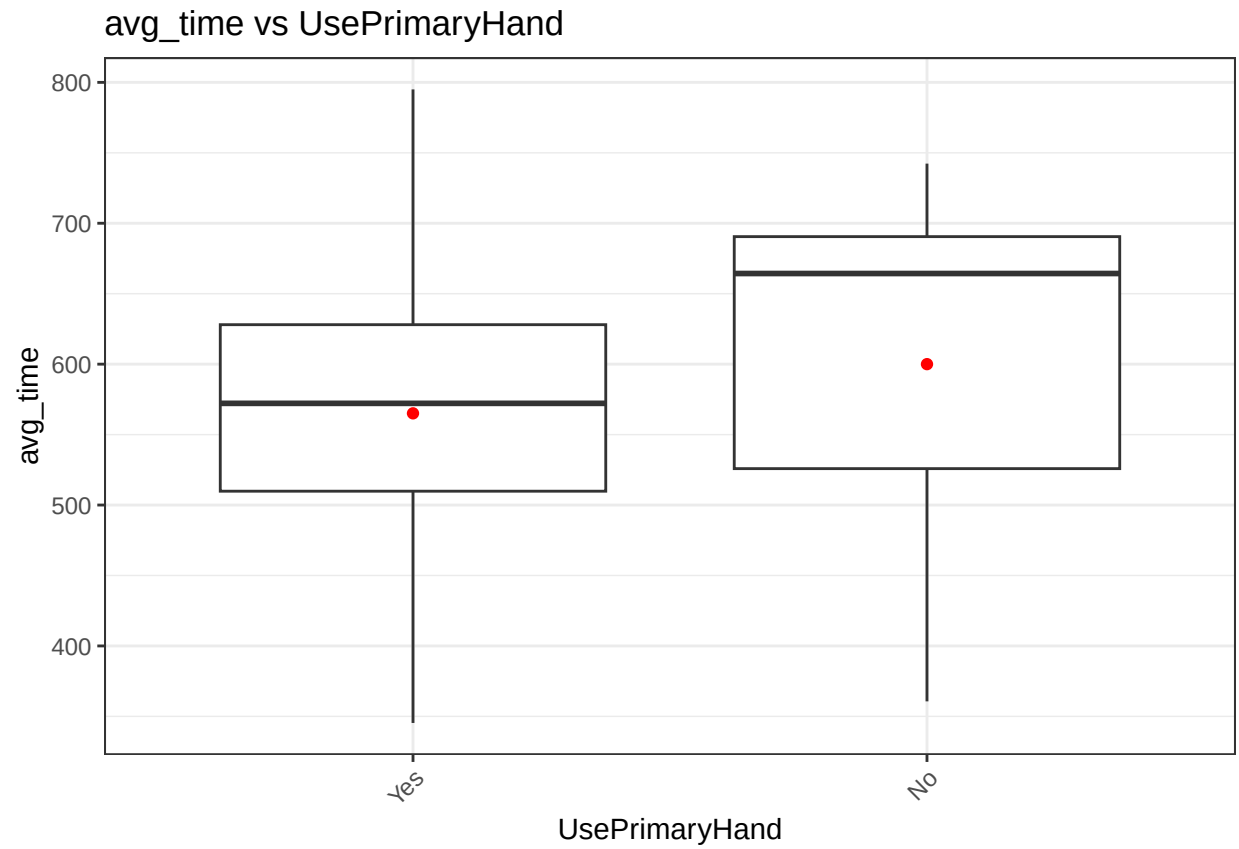
```
## avg_time vs InputDevice  
plot_box("InputDevice")
```



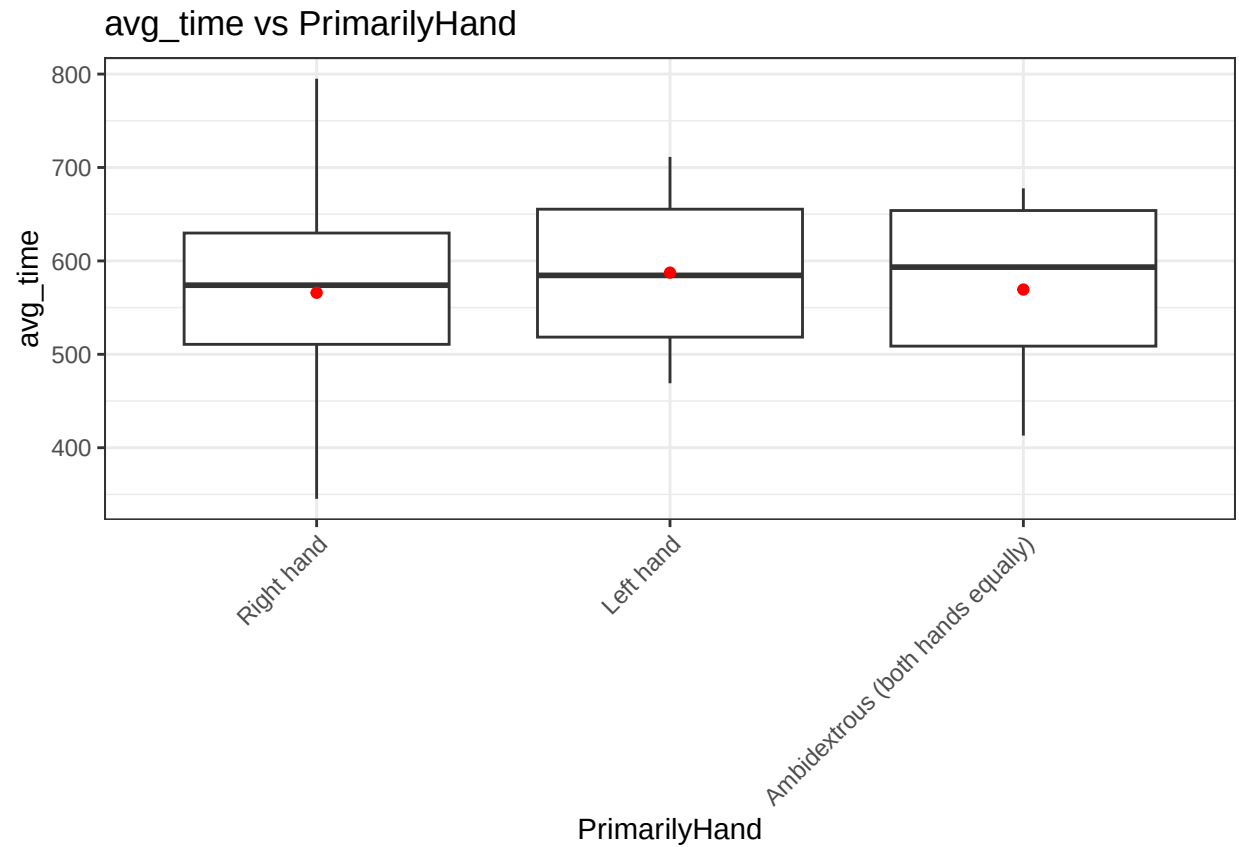
```
## avg_time vs Cautious  
plot_box("Cautious")
```



```
## avg_time vs UsePrimaryHand  
plot_box("UsePrimaryHand")
```

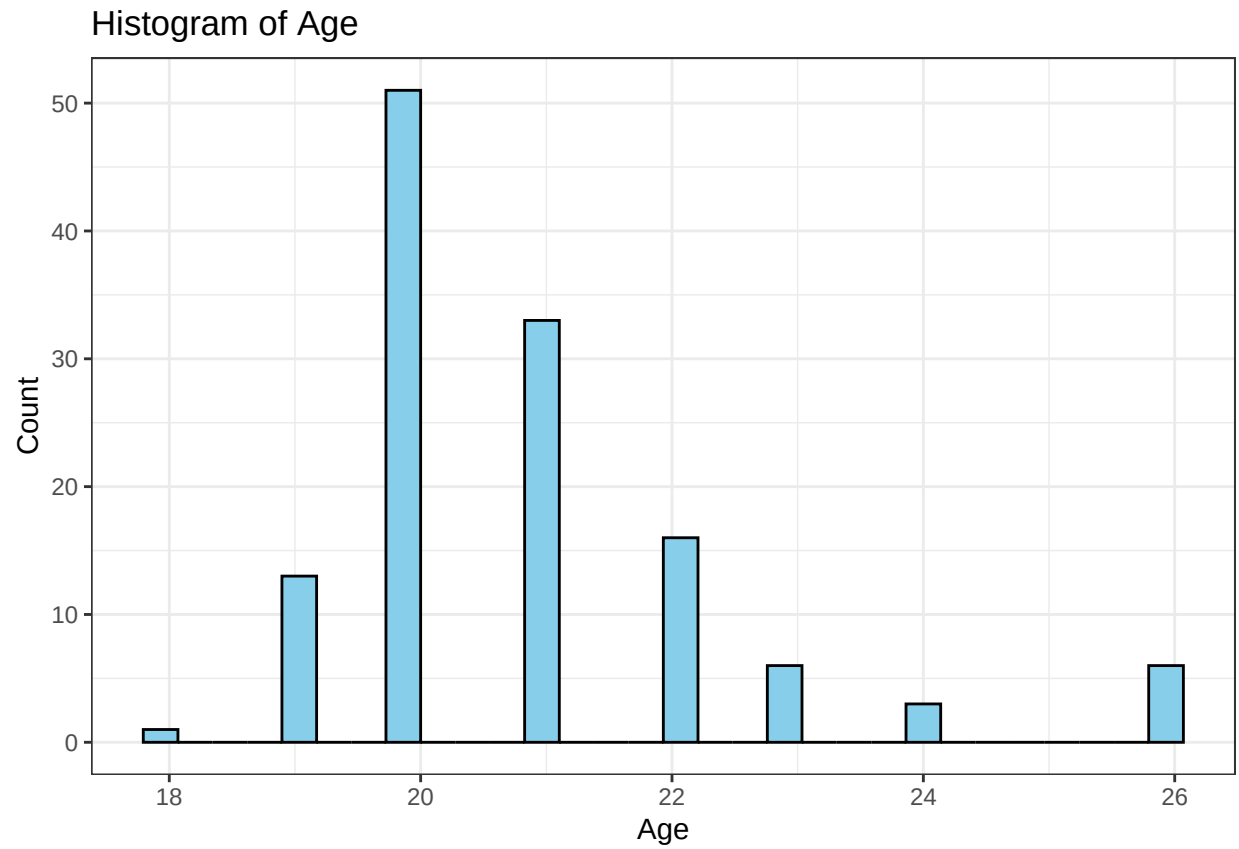


```
## avg_time vs PrimarilyHand  
plot_box("PrimarilyHand")
```

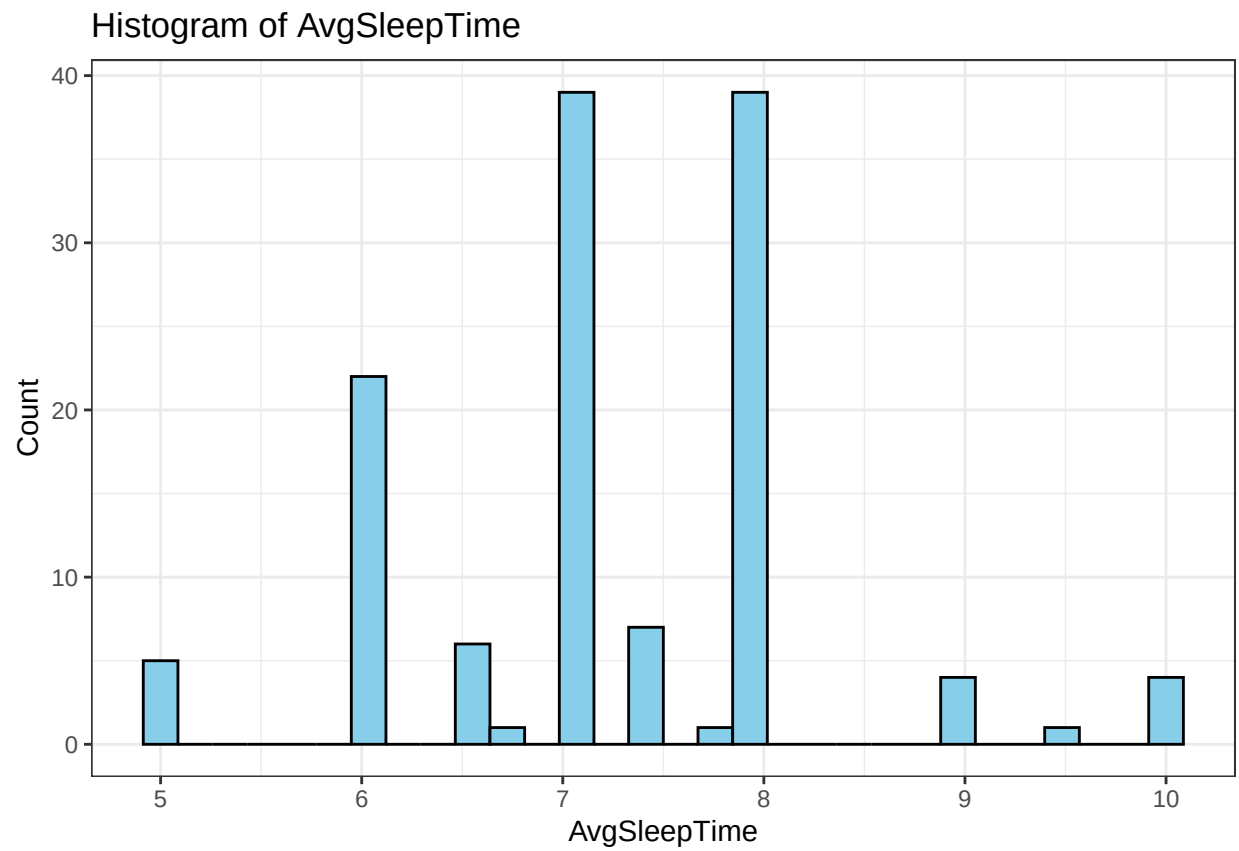


```
## Continuous variables histogram
plot_hist <- function(var, bins = 30) {
  survey_cleaned %>%
    ggplot(aes(x = {{ var }})) +
    geom_histogram(fill = "skyblue", color = "black", bins = bins) +
    labs(title = paste("Histogram of", deparse(substitute(var))),
         x = deparse(substitute(var)), y = "Count") +
    theme_bw()
}

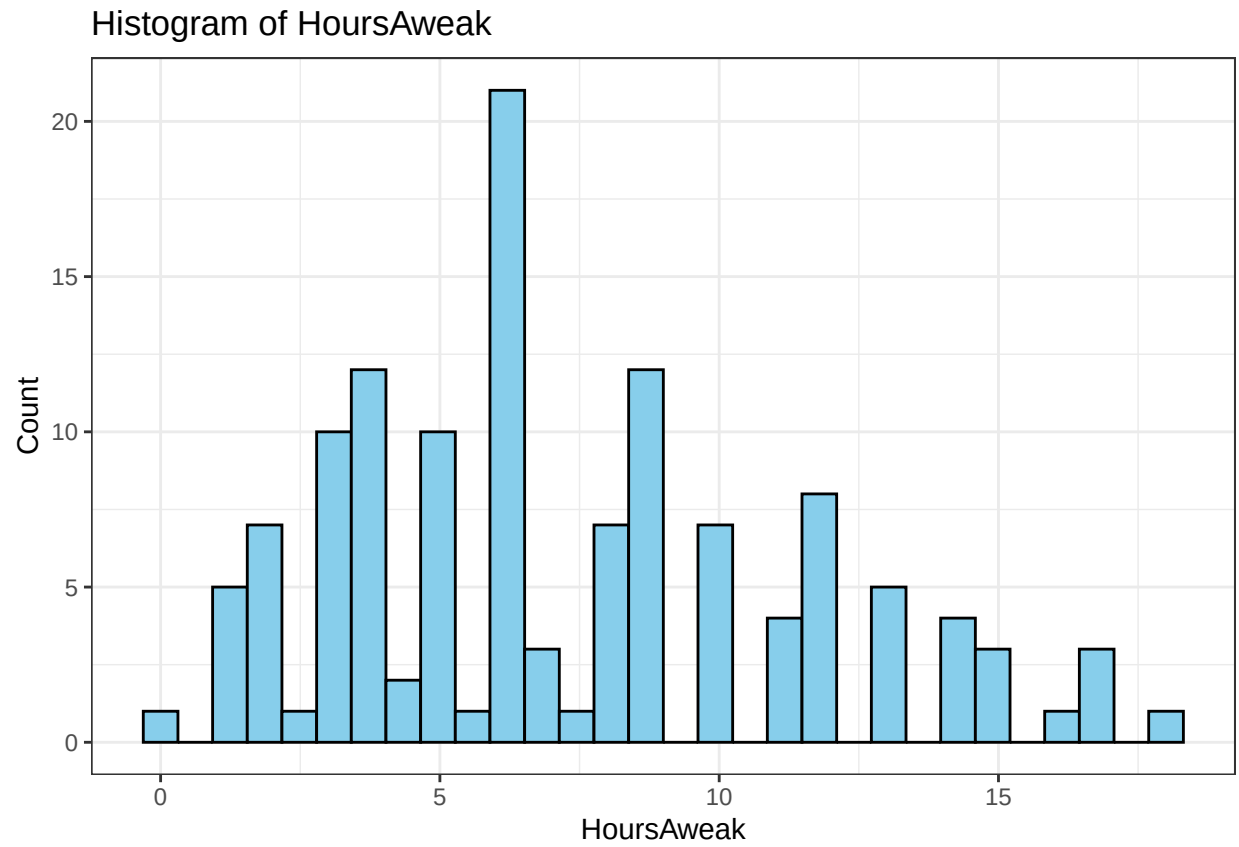
## Age
plot_hist(Age)
```



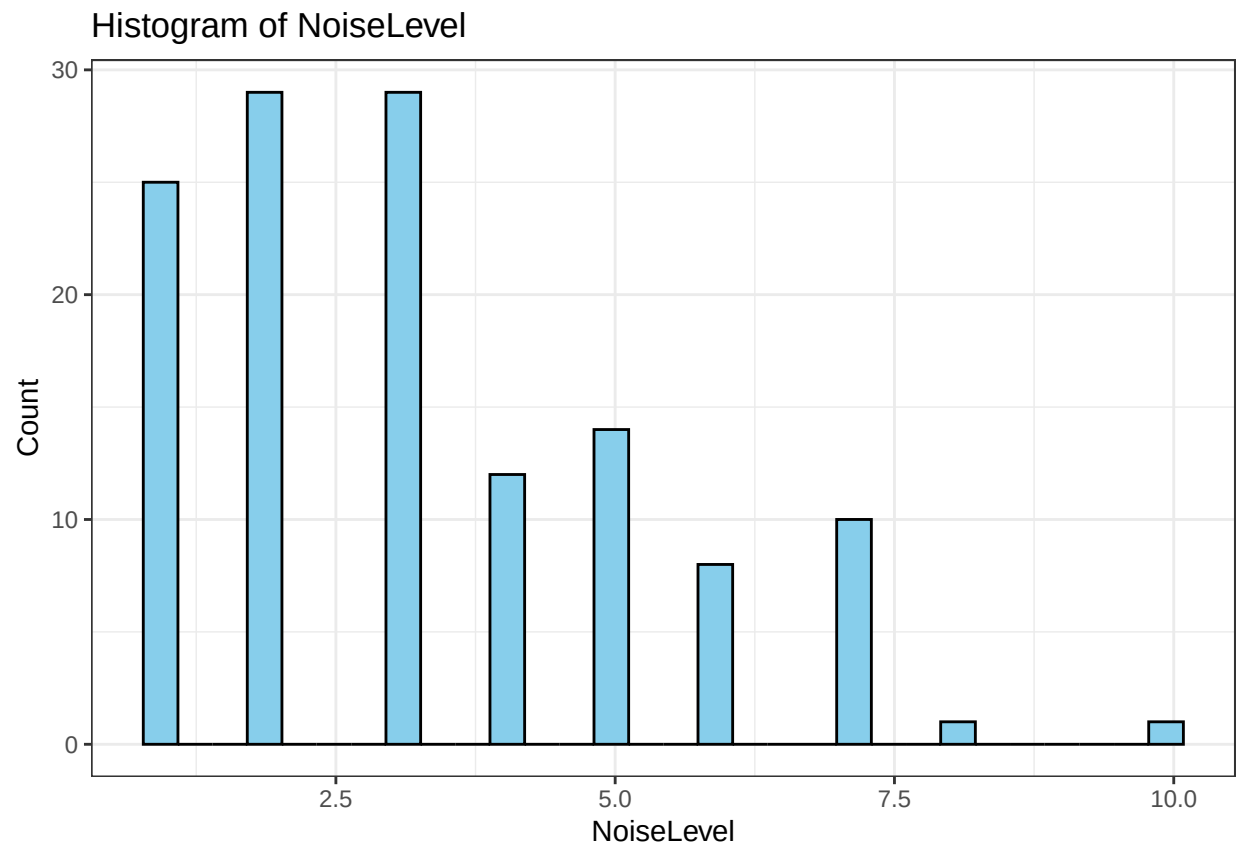
```
## AvgSleepTime  
plot_hist(AvgSleepTime)
```

```
## HoursAweek  
plot_hist(HoursAweek)
```

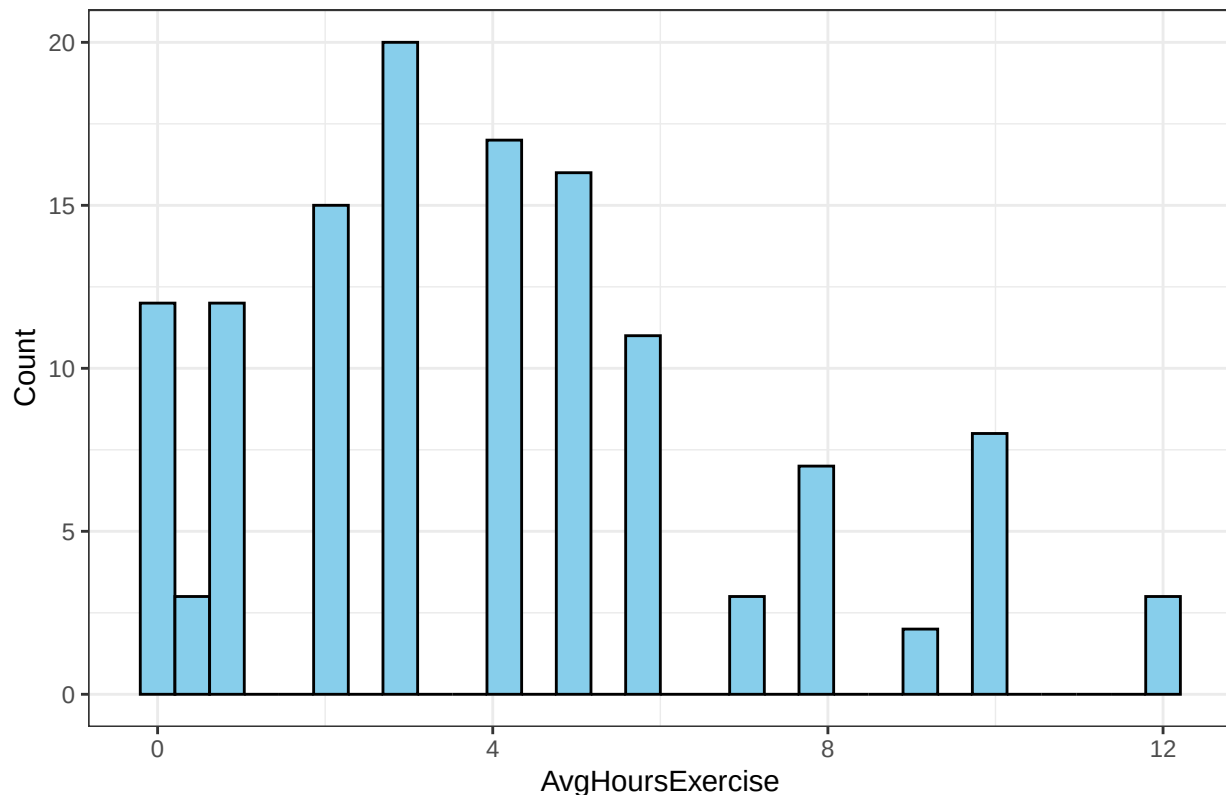


```
## NoiseLevel  
plot_hist(NoiseLevel)
```



```
## AvgHoursExercise  
plot_hist(AvgHoursExercise)
```

Histogram of AvgHoursExercise



```
# Pre-processing
Survey_cleaned <- survey_cleaned
## Combine 5 fatigue levels to 3 levels
Survey_cleaned <- Survey_cleaned %>%
  mutate(Fatigue_3 = case_when(
    Fatigue %in% c("Extremely fatigued", "Very fatigued") ~ "High Fatigue",
    Fatigue %in% c("Moderately fatigued") ~ "Moderately Fatigue",
    Fatigue %in% c("Not fatigued at all", "Slightly Fatigued") ~ "Low Fatigue",
    Fatigue_3 = factor(Fatigue_3, levels = c("High Fatigue", "Moderately Fatigue", "Low Fatigue"))))

## Combine 5 Stress levels to 3 levels
Survey_cleaned <- Survey_cleaned %>%
  mutate(Stress_3 = case_when(
    Stress %in% c("Very High", "High") ~ "High Stress",
    Stress %in% c("Moderate") ~ "Moderately Stress",
    Stress %in% c("Low", "Very Low") ~ "Low Stress",
    Stress_3 = factor(Stress_3, levels = c("High Stress", "Moderately Stress", "Low Stress"))))

## Combine 10 Noise levels to 3 levels
Survey_cleaned <- Survey_cleaned %>%
  mutate(Noise_Level = case_when(
    NoiseLevel %in% c(1,2,3) ~ "Low Noise",
    NoiseLevel %in% c(4,5,6,7) ~ "Moderately Noise",
    NoiseLevel %in% c(8,9,10) ~ "High Noise",
    Noise_Level = factor(Noise_Level, levels = c("High Noise", "Moderately Noise", "Low Noise"))))
```

```

## Combine 6 VideoGamePlay levels to 2 levels
Survey_cleaned <- Survey_cleaned %>%
  mutate(Game_Frequency = case_when(
    VideoGamePlay %in% c("Daily", "Several times a week", "Once a week") ~ "High Frequency",
    VideoGamePlay %in% c("Several times a month", "Rarely", "Never") ~ "Low Frequency",
    Game_Frequency = factor(Game_Frequency, levels = c("High Frequency", "Low Frequency")))

## Combine 6 SportsPlay levels to 2 levels
Survey_cleaned <- Survey_cleaned %>%
  mutate(Sports_Frequency = case_when(
    SportsPlay %in% c("Daily", "Several times a week", "Once a week") ~ "High Frequency",
    SportsPlay %in% c("Several times a month", "Rarely", "Never") ~ "Low Frequency",
    Sports_Frequency = factor(Sports_Frequency, levels = c("High Frequency", "Low Frequency")))

## Combine 5 ShootingGames levels to 3 levels
Survey_cleaned <- Survey_cleaned %>%
  mutate(Shooting_Levels = case_when(
    ShootingGames %in% c("Expert", "Advanced") ~ "High Levels",
    ShootingGames %in% c("Intermediate") ~ "Intermediate Levels",
    ShootingGames %in% c("Beginner", "None") ~ "Low Levels",
    Shooting_Levels = factor(Shooting_Levels, levels = c("High Levels",
                                                         "Intermediate Levels", "Low Levels")))

Survey_cleaned <- separate(Survey_cleaned, col = DeviceOS, into = c("Device", "OS"), sep = "-")

## Combine IOS and MacOS as Apple
Survey_cleaned <- Survey_cleaned %>%
  mutate(System = case_when(
    OS %in% c("IOS", "macOS") ~ "Apple",
    OS %in% "Windows" ~ "Windows",
    OS %in% "Linux" ~ "Linux"),
  System = factor(System))

## Transformation
Survey_cleaned$logAvgHoursExercise <- log(Survey_cleaned$AvgHoursExercise + 1)

## Remove not informative/used columns
Survey_cleaned <- Survey_cleaned %>%
  dplyr::select(-First) %>%
  dplyr::select(-Second) %>%
  dplyr::select(-Third) %>%
  dplyr::select(-Fatigue) %>%
  dplyr::select(-NoiseLevel) %>%
  dplyr::select(-VideoGamePlay) %>%
  dplyr::select(-SportsPlay) %>%
  dplyr::select(-Stress) %>%
  dplyr::select(-OS) %>%
  dplyr::select(-ShootingGames) %>%
  dplyr::select(-AvgHoursExercise)

# Fit the full model
full_model <- lm(avg_time ~ ., data = Survey_cleaned)
summary(full_model)

```

```
##
## Call:
## lm(formula = avg_time ~ ., data = Survey_cleaned)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -119.584  -31.677   -1.699   23.259  158.041
##
## Coefficients:
##                                     Estimate Std. Error t value
## (Intercept)                      316.7155    214.0247   1.480
## ClassJunior                       36.3028     19.3107   1.880
## ClassSenior                        5.9146     27.0350   0.219
## ClassGraduate student            120.6718     55.3700   2.179
## Age                               3.0620      8.1732   0.375
## AvgSleepTime                     15.7785      8.3699   1.885
## LastNightSleep                    1.7912      5.3886   0.332
## HoursAweak                        0.5274      2.0285   0.260
## DistractionYes                    28.4336     20.0834   1.416
## TemperatureWarm                   25.1289     64.2955   0.391
## TemperatureNeutral                46.3851     60.6552   0.765
## TemperatureCold                   65.9307     63.1520   1.044
## GamerYes                           15.4718     18.0549   0.857
## CaffeinIntakeYes                  40.1718     17.9718   2.235
## VisualAcuityExcellent              -58.0343     29.0626  -1.997
## VisualAcuityGood                  -79.1387     29.9986  -2.638
## VisualAcuityPoor                  -72.0108     50.5460  -1.425
## VisualAcuityVery Poor            -146.9141     97.1368  -1.512
## PrimarilyHandLeft hand            -54.6740     44.0925  -1.240
## PrimarilyHandAmbidextrous (both hands equally)  34.9272     40.1681   0.870
## UsePrimaryHandNo                  65.8135     40.6695   1.618
## CautiousVery cautious              3.0895     45.3541   0.068
## CautiousModerately cautious       -6.5308     46.0642  -0.142
## CautiousSlightly cautious          9.0990     48.4746   0.188
## CautiousNot cautious at all        38.3024     64.5652   0.593
## InputDeviceMouse                  -45.1659     78.2158  -0.577
## InputDeviceKeyboard                77.1231     96.5755   0.799
## InputDeviceTrackpad                37.3924     77.5345   0.482
## InputDeviceTouch screen            18.3343     85.9597   0.213
## RefreshRateMid-Range               30.3461     24.3793   1.245
## RefreshRateStandard                59.9600     22.9308   2.615
## DeviceiPad(Tablet                 -42.5427     64.8587  -0.656
## DeviceiPhone(Smartphone            81.0134     83.1190   0.975
## DeviceLaptop                       36.2946     24.0861   1.507
## WiFiStable                       -96.2096     58.0745  -1.657
## Fatigue_3Moderately Fatigue        54.4543     26.0192   2.093
## Fatigue_3Low Fatigue               48.2829     26.9053   1.795
## Stress_3Moderately Stress           14.0792     20.5947   0.684
## Stress_3Low Stress                 -29.3721     21.5548  -1.363
## Noise_LevelModerately Noise        -35.7234     71.8799  -0.497
## Noise_LevelLow Noise               -19.1215     71.4386  -0.268
## Game_FrequencyLow Frequency        21.8759     15.1069   1.448
## Sports_FrequencyLow Frequency       17.6599     18.2598   0.967
## Shooting_LevelsIntermediate Levels  -7.1078     24.4294  -0.291
```

## Shooting_LevelsLow Levels	21.0557	22.7596	0.925
## SystemLinux	39.9766	67.0464	0.596
## SystemWindows	-0.8867	19.2729	-0.046
## logAvgHoursExercise	3.1802	12.8470	0.248
##	Pr(> t)		
## (Intercept)	0.1428		
## ClassJunior	0.0637	.	
## ClassSenior	0.8274		
## ClassGraduate student	0.0322	*	
## Age	0.7089		
## AvgSleepTime	0.0630	.	
## LastNightSleep	0.7404		
## HoursAweak	0.7955		
## DistractionYes	0.1607		
## TemperatureWarm	0.6969		
## TemperatureNeutral	0.4467		
## TemperatureCold	0.2996		
## GamerYes	0.3940		
## CaffeinIntakeYes	0.0282	*	
## VisualAcuityExcellent	0.0492	*	
## VisualAcuityGood	0.0100	**	
## VisualAcuityPoor	0.1581		
## VisualAcuityVery Poor	0.1343		
## PrimarilyHandLeft hand	0.2186		
## PrimarilyHandAmbidextrous (both hands equally)	0.3871		
## UsePrimaryHandNo	0.1095		
## CautiousVery cautious	0.9459		
## CautiousModerately cautious	0.8876		
## CautiousSlightly cautious	0.8516		
## CautiousNot cautious at all	0.5547		
## InputDeviceMouse	0.5652		
## InputDeviceKeyboard	0.4269		
## InputDeviceTrackpad	0.6309		
## InputDeviceTouch screen	0.8316		
## RefreshRateMid-Range	0.2168		
## RefreshRateStandard	0.0106	*	
## DeviceiPad(Tablet	0.5137		
## DeviceiPhone(Smartphone	0.3326		
## DeviceLaptop	0.1357		
## WiFiStable	0.1015		
## Fatigue_3Moderately Fatigue	0.0395	*	
## Fatigue_3Low Fatigue	0.0765	.	
## Stress_3Moderately Stress	0.4962		
## Stress_3Low Stress	0.1768		
## Noise_LevelModerately Noise	0.6205		
## Noise_LevelLow Noise	0.7896		
## Game_FrequencyLow Frequency	0.1515		
## Sports_FrequencyLow Frequency	0.3364		
## Shooting_LevelsIntermediate Levels	0.7718		
## Shooting_LevelsLow Levels	0.3576		
## SystemLinux	0.5527		
## SystemWindows	0.9634		
## logAvgHoursExercise	0.8051		
## ---			

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 66.77 on 81 degrees of freedom
## Multiple R-squared:  0.6914, Adjusted R-squared:  0.5123
## F-statistic: 3.861 on 47 and 81 DF,  p-value: 4.833e-08
```

```
# Variable Selection
## Searching methods AIC stepwise
step(full_model, direction = "both")
```

```
## Start:  AIC=1119.88
## avg_time ~ Class + Age + AvgSleepTime + LastNightSleep + HoursAweak +
##   Distraction + Temperature + Gamer + CaffeinIntake + VisualAcuity +
##   PrimarilyHand + UsePrimaryHand + Cautious + InputDevice +
##   RefreshRate + Device + WiFi + Fatigue_3 + Stress_3 + Noise_Level +
##   Game_Frequency + Sports_Frequency + Shooting_Levels + System +
##   logAvgHoursExercise
```

	Df	Sum of Sq	RSS	AIC
- Cautious	4	6186	367259	1114.1
- System	2	1980	363052	1116.6
- Noise_Level	2	5074	366147	1117.7
- Temperature	3	10845	371918	1117.7
- logAvgHoursExercise	1	273	361346	1118.0
- HoursAweak	1	301	361374	1118.0
- LastNightSleep	1	493	361565	1118.0
- Age	1	626	361699	1118.1
- Gamer	1	3273	364346	1119.0
- Sports_Frequency	1	4170	365243	1119.4
<none>			361073	1119.9
- PrimarilyHand	2	11526	372599	1119.9
- Shooting_Levels	2	11896	372969	1120.1
- Distraction	1	8935	370008	1121.0
- Game_Frequency	1	9347	370420	1121.2
- Device	3	22755	383828	1121.8
- UsePrimaryHand	1	11674	372746	1122.0
- WiFi	1	12234	373307	1122.2
- Fatigue_3	2	19740	380813	1122.7
- AvgSleepTime	1	15842	376915	1123.4
- Stress_3	2	23328	384401	1124.0
- VisualAcuity	4	37145	398218	1124.5
- CaffeinIntake	1	22272	383345	1125.6
- Class	3	40144	401217	1127.5
- RefreshRate	2	34921	395994	1127.8
- InputDevice	4	73291	434364	1135.7

```
## Step:  AIC=1114.07
## avg_time ~ Class + Age + AvgSleepTime + LastNightSleep + HoursAweak +
##   Distraction + Temperature + Gamer + CaffeinIntake + VisualAcuity +
##   PrimarilyHand + UsePrimaryHand + InputDevice + RefreshRate +
##   Device + WiFi + Fatigue_3 + Stress_3 + Noise_Level + Game_Frequency +
##   Sports_Frequency + Shooting_Levels + System + logAvgHoursExercise
##
##           Df Sum of Sq  RSS  AIC
```



```

## - System                2      3458 370717 1111.3
## - Temperature           3     11208 378467 1112.0
## - logAvgHoursExercise   1         0 367259 1112.1
## - Age                   1       450 367708 1112.2
## - LastNightSleep        1       519 367778 1112.2
## - HoursAweak            1       611 367869 1112.3
## - Noise_Level           2     8204 375463 1112.9
## - Gamer                 1     3617 370875 1113.3
## - Sports_Frequency      1     4109 371368 1113.5
## - PrimarilyHand         2    10345 377604 1113.7
## <none>                  367259 1114.1
## - Shooting_Levels       2    14422 381681 1115.0
## - Distraction           1    10066 377324 1115.6
## - Device                3    22527 389786 1115.8
## - Game_Frequency        1    10895 378153 1115.8
## - WiFi                 1    12477 379736 1116.4
## - UsePrimaryHand        1    15475 382733 1117.4
## - Fatigue_3             2    23869 391128 1118.2
## - Stress_3              2    23955 391213 1118.2
## - VisualAcuity          4    39758 407017 1119.3
## - CaffeinIntake         1    22391 389650 1119.7
## + Cautious              4     6186 361073 1119.9
## - AvgSleepTime          1    26198 393457 1121.0
## - Class                 3    40028 407286 1121.4
## - RefreshRate           2    36741 404000 1122.4
## - InputDevice           4    74330 441589 1129.8
##
## Step:  AIC=1111.28
## avg_time ~ Class + Age + AvgSleepTime + LastNightSleep + HoursAweak +
##   Distraction + Temperature + Gamer + CaffeinIntake + VisualAcuity +
##   PrimarilyHand + UsePrimaryHand + InputDevice + RefreshRate +
##   Device + WiFi + Fatigue_3 + Stress_3 + Noise_Level + Game_Frequency +
##   Sports_Frequency + Shooting_Levels + logAvgHoursExercise
##
##              Df Sum of Sq    RSS    AIC
## - logAvgHoursExercise  1      18 370734 1109.3
## - HoursAweak           1      91 370807 1109.3
## - Temperature          3    11934 382651 1109.4
## - Noise_Level          2     6813 377529 1109.6
## - Age                  1     1077 371794 1109.7
## - LastNightSleep       1     1080 371797 1109.7
## - Gamer                1     4380 375097 1110.8
## - PrimarilyHand        2    10668 381384 1110.9
## - Sports_Frequency     1     5524 376240 1111.2
## <none>                 370717 1111.3
## - Distraction          1     9422 380139 1112.5
## - Shooting_Levels      2    15720 386437 1112.6
## - Game_Frequency       1    10570 381286 1112.9
## - Device               3    23286 394002 1113.1
## - WiFi                 1    13732 384449 1114.0
## + System               2     3458 367259 1114.1
## - Fatigue_3            2    21150 391867 1114.4
## - UsePrimaryHand       1    16142 386858 1114.8
## - Stress_3             2    26900 397616 1116.3

```

```

## + Cautious          4      7664 363052 1116.6
## - VisualAcuity      4      40603 411320 1116.7
## - CaffeinIntake     1      23349 394066 1117.2
## - AvgSleepTime      1      23564 394280 1117.2
## - Class              3      37904 408621 1117.8
## - RefreshRate       2      33438 404154 1118.4
## - InputDevice       4      105407 476123 1135.6
##
## Step: AIC=1109.28
## avg_time ~ Class + Age + AvgSleepTime + LastNightSleep + HoursAweak +
##   Distraction + Temperature + Gamer + CaffeinIntake + VisualAcuity +
##   PrimarilyHand + UsePrimaryHand + InputDevice + RefreshRate +
##   Device + WiFi + Fatigue_3 + Stress_3 + Noise_Level + Game_Frequency +
##   Sports_Frequency + Shooting_Levels
##
##              Df Sum of Sq    RSS    AIC
## - HoursAweak      1         87 370821 1107.3
## - Temperature     3      11961 382695 1107.4
## - Noise_Level      2       6820 377555 1107.6
## - Age              1       1060 371794 1107.7
## - LastNightSleep   1       1064 371798 1107.7
## - Gamer            1       4366 375100 1108.8
## - PrimarilyHand    2      10827 381561 1109.0
## <none>              370734 1109.3
## - Sports_Frequency 1       5919 376653 1109.3
## - Distraction      1       9495 380229 1110.5
## - Shooting_Levels  2     16442 387176 1110.9
## - Game_Frequency   1     10647 381381 1110.9
## - Device           3     23274 394008 1111.1
## + logAvgHoursExercise 1        18 370717 1111.3
## - WiFi             1     13743 384477 1112.0
## + System           2       3475 367259 1112.1
## - Fatigue_3        2     21828 392562 1112.7
## - UsePrimaryHand    1     16130 386865 1112.8
## - Stress_3         2     26889 397623 1114.3
## + Cautious         4       7260 363474 1114.7
## - VisualAcuity     4     40924 411658 1114.8
## - AvgSleepTime     1     23825 394559 1115.3
## - CaffeinIntake    1     24115 394849 1115.4
## - Class            3     37930 408664 1115.8
## - RefreshRate      2     34559 405293 1116.8
## - InputDevice      4     105750 476484 1133.7
##
## Step: AIC=1107.31
## avg_time ~ Class + Age + AvgSleepTime + LastNightSleep + Distraction +
##   Temperature + Gamer + CaffeinIntake + VisualAcuity + PrimarilyHand +
##   UsePrimaryHand + InputDevice + RefreshRate + Device + WiFi +
##   Fatigue_3 + Stress_3 + Noise_Level + Game_Frequency + Sports_Frequency +
##   Shooting_Levels
##
##              Df Sum of Sq    RSS    AIC
## - Temperature     3     11970 382791 1105.4
## - Noise_Level      2       6813 377635 1105.7
## - LastNightSleep   1       1066 371888 1105.7

```

```

## - Age 1 1105 371926 1105.7
## - Gamer 1 4474 375295 1106.9
## <none> 370821 1107.3
## - Sports_Frequency 1 6209 377030 1107.5
## - PrimarilyHand 2 12830 383651 1107.7
## - Distraction 1 9417 380238 1108.5
## - Game_Frequency 1 10560 381381 1108.9
## - Shooting_Levels 2 16837 387659 1109.0
## - Device 3 23213 394034 1109.2
## + HoursAweak 1 87 370734 1109.3
## + logAvgHoursExercise 1 14 370807 1109.3
## - WiFi 1 13749 384571 1110.0
## + System 2 2952 367870 1110.3
## - Fatigue_3 2 23362 394183 1111.2
## - UsePrimaryHand 1 17655 388477 1111.3
## - Stress_3 2 27142 397963 1112.4
## + Cautious 4 7320 363501 1112.7
## - VisualAcuity 4 41189 412010 1112.9
## - AvgSleepTime 1 23826 394648 1113.3
## - CaffeinIntake 1 24468 395289 1113.6
## - Class 3 38459 409280 1114.0
## - RefreshRate 2 34613 405434 1114.8
## - InputDevice 4 107672 478493 1132.2
##
## Step: AIC=1105.41
## avg_time ~ Class + Age + AvgSleepTime + LastNightSleep + Distraction +
##   Gamer + CaffeinIntake + VisualAcuity + PrimarilyHand + UsePrimaryHand +
##   InputDevice + RefreshRate + Device + WiFi + Fatigue_3 + Stress_3 +
##   Noise_Level + Game_Frequency + Sports_Frequency + Shooting_Levels
##
##           Df Sum of Sq    RSS    AIC
## - Noise_Level 2 4281 387073 1102.8
## - LastNightSleep 1 284 383075 1103.5
## - Age 1 1279 384070 1103.8
## - Gamer 1 1827 384618 1104.0
## - Sports_Frequency 1 5120 387912 1105.1
## <none> 382791 1105.4
## - Shooting_Levels 2 12603 395394 1105.6
## - Distraction 1 8186 390978 1106.1
## - Game_Frequency 1 8210 391001 1106.2
## - PrimarilyHand 2 16490 399282 1106.8
## - Device 3 23846 406638 1107.2
## + Temperature 3 11970 370821 1107.3
## + HoursAweak 1 97 382695 1107.4
## + logAvgHoursExercise 1 40 382751 1107.4
## + System 2 3673 379119 1108.2
## - WiFi 1 15089 397880 1108.4
## - Stress_3 2 23613 406404 1109.1
## - VisualAcuity 4 36566 419357 1109.2
## - Fatigue_3 2 28179 410971 1110.6
## - UsePrimaryHand 1 21981 404773 1110.6
## + Cautious 4 7923 374869 1110.7
## - AvgSleepTime 1 23256 406048 1111.0
## - Class 3 36590 419382 1111.2

```

```

## - CaffeinIntake      1      25129 407921 1111.6
## - RefreshRate       2      36072 418863 1113.0
## - InputDevice       4      100467 483258 1127.5
##
## Step: AIC=1102.85
## avg_time ~ Class + Age + AvgSleepTime + LastNightSleep + Distraction +
##      Gamer + CaffeinIntake + VisualAcuity + PrimarilyHand + UsePrimaryHand +
##      InputDevice + RefreshRate + Device + WiFi + Fatigue_3 + Stress_3 +
##      Game_Frequency + Sports_Frequency + Shooting_Levels
##
##              Df Sum of Sq    RSS    AIC
## - LastNightSleep      1         383 387456 1101.0
## - Gamer                1         1318 388391 1101.3
## - Age                  1         1614 388687 1101.4
## - Shooting_Levels     2        11193 398266 1102.5
## <none>                  387073 1102.8
## - Distraction          1         6132 393205 1102.9
## - Sports_Frequency     1         6408 393481 1103.0
## - Game_Frequency       1         8205 395278 1103.5
## - PrimarilyHand        2        16349 403422 1104.2
## + HoursAweak           1          74 386999 1104.8
## + logAvgHoursExercise  1          1 387071 1104.8
## + Noise_Level          2         4281 382791 1105.4
## - WiFi                 1        14680 401753 1105.7
## + Temperature          3         9438 377635 1105.7
## + System                2         2576 384497 1106.0
## - Stress_3             2        23485 410557 1106.4
## - Device               3        31250 418322 1106.9
## - VisualAcuity         4        38848 425921 1107.2
## + Cautious             4         9580 377493 1107.6
## - Fatigue_3            2        27266 414338 1107.6
## - Class                3        34189 421262 1107.8
## - AvgSleepTime         1        23007 410080 1108.3
## - CaffeinIntake        1        23850 410922 1108.6
## - UsePrimaryHand       1        24556 411629 1108.8
## - RefreshRate          2        35695 422768 1110.2
## - InputDevice          4        98624 485696 1124.1
##
## Step: AIC=1100.97
## avg_time ~ Class + Age + AvgSleepTime + Distraction + Gamer +
##      CaffeinIntake + VisualAcuity + PrimarilyHand + UsePrimaryHand +
##      InputDevice + RefreshRate + Device + WiFi + Fatigue_3 + Stress_3 +
##      Game_Frequency + Sports_Frequency + Shooting_Levels
##
##              Df Sum of Sq    RSS    AIC
## - Gamer                1         1334 388790 1099.4
## - Age                  1         1828 389284 1099.6
## - Shooting_Levels     2        11178 398633 1100.6
## - Distraction          1         5951 393406 1100.9
## <none>                  387456 1101.0
## - Sports_Frequency     1         6316 393772 1101.1
## - Game_Frequency       1         8139 395594 1101.7
## - PrimarilyHand        2        16257 403712 1102.3
## + LastNightSleep      1          383 387073 1102.8

```

```

## + HoursAweak          1          76 387380 1103.0
## + logAvgHoursExercise  1          17 387438 1103.0
## + Noise_Level         2         4380 383075 1103.5
## - WiFi                1        14370 401826 1103.7
## + Temperature         3         8701 378754 1104.0
## + System              2         2691 384764 1104.1
## - Stress_3            2        23111 410566 1104.5
## - Device              3        31664 419119 1105.1
## - VisualAcuity        4        38508 425964 1105.2
## + Cautious            4         9643 377812 1105.7
## - Class               3        34943 422398 1106.1
## - CaffeinIntake       1        23961 411416 1106.7
## - Fatigue_3           2        30543 417998 1106.8
## - UsePrimaryHand      1        24445 411900 1106.9
## - RefreshRate         2        35509 422965 1108.3
## - AvgSleepTime        1        29291 416746 1108.4
## - InputDevice         4       100671 488127 1122.8
##
## Step:  AIC=1099.42
## avg_time ~ Class + Age + AvgSleepTime + Distraction + CaffeinIntake +
##      VisualAcuity + PrimarilyHand + UsePrimaryHand + InputDevice +
##      RefreshRate + Device + WiFi + Fatigue_3 + Stress_3 + Game_Frequency +
##      Sports_Frequency + Shooting_Levels
##
##              Df Sum of Sq    RSS    AIC
## - Age          1      2399 391188 1098.2
## - Shooting_Levels  2     10171 398960 1098.8
## - Distraction    1      5525 394315 1099.2
## - Sports_Frequency 1      5816 394606 1099.3
## <none>                      388790 1099.4
## - Game_Frequency  1      6920 395710 1099.7
## - PrimarilyHand   2     15844 404634 1100.6
## + Gamer           1      1334 387456 1101.0
## + LastNightSleep  1       399 388391 1101.3
## + HoursAweak      1       123 388667 1101.4
## + logAvgHoursExercise 1       55 388734 1101.4
## - WiFi            1     13216 402006 1101.7
## + Noise_Level     2       3866 384924 1102.1
## + System          2       2804 385985 1102.5
## - Stress_3        2     22081 410871 1102.5
## + Temperature     3       6969 381821 1103.1
## - VisualAcuity    4     37179 425969 1103.2
## - Device          3     31474 420264 1103.5
## + Cautious        4       9881 378909 1104.1
## - Class           3     33671 422461 1104.1
## - CaffeinIntake   1     22733 411522 1104.8
## - Fatigue_3       2     29498 418288 1104.8
## - UsePrimaryHand  1     23127 411916 1104.9
## - AvgSleepTime    1     28824 417614 1106.6
## - RefreshRate     2     35662 424452 1106.7
## - InputDevice     4     100536 489325 1121.1
##
## Step:  AIC=1098.21
## avg_time ~ Class + AvgSleepTime + Distraction + CaffeinIntake +

```

```

## VisualAcuity + PrimarilyHand + UsePrimaryHand + InputDevice +
## RefreshRate + Device + WiFi + Fatigue_3 + Stress_3 + Game_Frequency +
## Sports_Frequency + Shooting_Levels
##
##          Df Sum of Sq    RSS    AIC
## - Shooting_Levels      2      9379 400568 1097.3
## - Distraction          1      5870 397058 1098.1
## <none>                  391188 1098.2
## - Sports_Frequency      1      7300 398489 1098.6
## - Game_Frequency        1      7549 398738 1098.7
## + Age                   1      2399 388790 1099.4
## + Gamer                 1      1905 389284 1099.6
## - PrimarilyHand         2     17520 408708 1099.9
## + LastNightSleep        1       661 390527 1100.0
## - WiFi                  1     11941 403130 1100.1
## + logAvgHoursExercise   1       275 390913 1100.1
## + HoursAweak            1       211 390977 1100.1
## + Noise_Level           2      4203 386986 1100.8
## + System                2      3594 387594 1101.0
## - Stress_3              2     22189 413377 1101.3
## - VisualAcuity          4     35439 426628 1101.4
## + Temperature           3      6681 384507 1102.0
## - Device                3     32199 423387 1102.4
## + Cautious              4     10431 380757 1102.7
## - Fatigue_3             2     30846 422034 1104.0
## - UsePrimaryHand        1     25826 417014 1104.5
## - CaffeinIntake         1     25935 417123 1104.5
## - AvgSleepTime          1     26988 418176 1104.8
## - RefreshRate           2     35935 427124 1105.5
## - Class                 3     80062 471250 1116.2
## - InputDevice           4     99795 490984 1119.5
##
## Step:  AIC=1097.27
## avg_time ~ Class + AvgSleepTime + Distraction + CaffeinIntake +
## VisualAcuity + PrimarilyHand + UsePrimaryHand + InputDevice +
## RefreshRate + Device + WiFi + Fatigue_3 + Stress_3 + Game_Frequency +
## Sports_Frequency
##
##          Df Sum of Sq    RSS    AIC
## - Distraction          1      5333 405900 1097.0
## <none>                  400568 1097.3
## - WiFi                 1      8805 409372 1098.1
## - PrimarilyHand        2     15495 416062 1098.2
## + Shooting_Levels      2      9379 391188 1098.2
## - Sports_Frequency      1     10001 410569 1098.5
## + Age                   1      1607 398960 1098.8
## + logAvgHoursExercise   1      1285 399283 1098.8
## + Gamer                 1       605 399962 1099.1
## + LastNightSleep        1       592 399976 1099.1
## + HoursAweak            1       476 400092 1099.1
## - Game_Frequency        1     12442 413010 1099.2
## + System                2      4113 396455 1099.9
## + Noise_Level           2      2983 397584 1100.3
## - Stress_3              2     24562 425129 1100.9

```

```

## + Cautious          4      11847 388721 1101.4
## + Temperature       3       4947 395620 1101.7
## - VisualAcuity      4      40553 441121 1101.7
## - UsePrimaryHand    1      20795 421363 1101.8
## - Device            3      37731 438298 1102.9
## - CaffeinIntake     1      25512 426080 1103.2
## - AvgSleepTime      1      25819 426386 1103.3
## - Fatigue_3         2      39403 439970 1105.4
## - RefreshRate       2      40891 441458 1105.8
## - Class             3      94721 495289 1118.7
## - InputDevice       4     111509 512077 1121.0
##
## Step: AIC=1096.97
## avg_time ~ Class + AvgSleepTime + CaffeinIntake + VisualAcuity +
##   PrimarilyHand + UsePrimaryHand + InputDevice + RefreshRate +
##   Device + WiFi + Fatigue_3 + Stress_3 + Game_Frequency + Sports_Frequency
##
##              Df Sum of Sq    RSS    AIC
## <none>                405900 1097.0
## - PrimarilyHand      2      13237 419137 1097.1
## + Distraction        1       5333 400568 1097.3
## - WiFi              1       8406 414306 1097.6
## - Sports_Frequency   1       8986 414886 1097.8
## + Shooting_Levels    2      8842 397058 1098.1
## + Age               1       1943 403957 1098.3
## + logAvgHoursExercise 1       1905 403995 1098.4
## - Game_Frequency     1     11911 417811 1098.7
## + LastNightSleep     1       428 405473 1098.8
## + HoursAweak         1       399 405501 1098.8
## + Gamer              1       398 405503 1098.8
## + System             2       3809 402091 1099.8
## + Noise_Level        2       1132 404768 1100.6
## + Cautious           4     12107 393793 1101.1
## + Temperature        3       4990 400910 1101.4
## - UsePrimaryHand     1     20798 426698 1101.4
## - Stress_3           2     28495 434395 1101.7
## - VisualAcuity       4     43516 449416 1102.1
## - AvgSleepTime       1     24693 430593 1102.6
## - Device            3     38320 444220 1102.6
## - CaffeinIntake      1     33563 439463 1105.2
## - Fatigue_3          2     41583 447483 1105.5
## - RefreshRate        2     42714 448614 1105.9
## - Class             3     93752 499652 1117.8
## - InputDevice       4    115093 520993 1121.2
##
##
## Call:
## lm(formula = avg_time ~ Class + AvgSleepTime + CaffeinIntake +
##   VisualAcuity + PrimarilyHand + UsePrimaryHand + InputDevice +
##   RefreshRate + Device + WiFi + Fatigue_3 + Stress_3 + Game_Frequency +
##   Sports_Frequency, data = Survey_cleaned)
##
## Coefficients:
##                               (Intercept)

```

```

##                438.055
##                ClassJunior
##                38.454
##                ClassSenior
##                15.073
##                ClassGraduate student
##                129.289
##                AvgSleepTime
##                15.025
##                CaffeinIntakeYes
##                43.014
##                VisualAcuityExcellent
##                -69.927
##                VisualAcuityGood
##                -83.490
##                VisualAcuityPoor
##                -73.772
##                VisualAcuityVery Poor
##                -116.392
##                PrimarilyHandLeft hand
##                -51.238
## PrimarilyHandAmbidextrous (both hands equally)
##                31.844
##                UsePrimaryHandNo
##                74.840
##                InputDeviceMouse
##                -51.908
##                InputDeviceKeyboard
##                53.504
##                InputDeviceTrackpad
##                25.861
##                InputDeviceTouch screen
##                -20.490
##                RefreshRateMid-Range
##                27.233
##                RefreshRateStandard
##                57.230
##                DeviceiPad(Tablet
##                -24.929
##                DeviceiPhone(Smartphone
##                142.767
##                DeviceLaptop
##                36.504
##                WiFiStable
##                -71.611
##                Fatigue_3Moderately Fatigue
##                63.388
##                Fatigue_3Low Fatigue
##                56.646
##                Stress_3Moderately Stress
##                7.428
##                Stress_3Low Stress
##                -32.981
##                Game_FrequencyLow Frequency

```



```
##                                21.879
##      Sports_FrequencyLow Frequency
##                                20.661
```

```
## Searching methods BIC stepwise
step(full_model, direction = "both", k = log(129))
```

```
## Start: AIC=1257.15
## avg_time ~ Class + Age + AvgSleepTime + LastNightSleep + HoursAweak +
##      Distraction + Temperature + Gamer + CaffeinIntake + VisualAcuity +
##      PrimarilyHand + UsePrimaryHand + Cautious + InputDevice +
##      RefreshRate + Device + WiFi + Fatigue_3 + Stress_3 + Noise_Level +
##      Game_Frequency + Sports_Frequency + Shooting_Levels + System +
##      logAvgHoursExercise
```

```
##
##      Df Sum of Sq    RSS    AIC
## - Cautious      4      6186 367259 1239.9
## - Temperature    3     10845 371918 1246.4
## - System         2      1980 363052 1248.1
## - Noise_Level    2      5074 366147 1249.2
## - VisualAcuity   4      37145 398218 1250.3
## - Device         3     22755 383828 1250.5
## - PrimarilyHand  2     11526 372599 1251.5
## - Shooting_Levels 2     11896 372969 1251.6
## - logAvgHoursExercise 1       273 361346 1252.4
## - HoursAweak     1       301 361374 1252.4
## - LastNightSleep 1       493 361565 1252.5
## - Age            1       626 361699 1252.5
## - Gamer          1      3273 364346 1253.5
## - Sports_Frequency 1      4170 365243 1253.8
## - Fatigue_3      2     19740 380813 1254.3
## - Distraction    1      8935 370008 1255.4
## - Stress_3       2     23328 384401 1255.5
## - Game_Frequency 1      9347 370420 1255.6
## - Class          3     40144 401217 1256.2
## - UsePrimaryHand 1     11674 372746 1256.4
## - WiFi           1     12234 373307 1256.6
## <none>                                361073 1257.2
## - AvgSleepTime   1     15842 376915 1257.8
## - RefreshRate     2     34921 395994 1259.3
## - CaffeinIntake   1     22272 383345 1260.0
## - InputDevice     4      73291 434364 1261.5
##
```

```
## Step: AIC=1239.9
## avg_time ~ Class + Age + AvgSleepTime + LastNightSleep + HoursAweak +
##      Distraction + Temperature + Gamer + CaffeinIntake + VisualAcuity +
##      PrimarilyHand + UsePrimaryHand + InputDevice + RefreshRate +
##      Device + WiFi + Fatigue_3 + Stress_3 + Noise_Level + Game_Frequency +
##      Sports_Frequency + Shooting_Levels + System + logAvgHoursExercise
##
##      Df Sum of Sq    RSS    AIC
## - Temperature    3     11208 378467 1229.2
## - System          2      3458 370717 1231.4
## - Device          3     22527 389786 1233.0
```

```

## - Noise_Level      2      8204 375463 1233.0
## - VisualAcuity     4      39758 407017 1233.7
## - PrimarilyHand    2      10345 377604 1233.8
## - logAvgHoursExercise 1        0 367259 1235.0
## - Shooting_Levels  2      14422 381681 1235.2
## - Age              1        450 367708 1235.2
## - LastNightSleep   1        519 367778 1235.2
## - HoursAweak       1        611 367869 1235.2
## - Gamer            1        3617 370875 1236.3
## - Sports_Frequency 1        4109 371368 1236.5
## - Fatigue_3        2      23869 391128 1238.3
## - Stress_3         2      23955 391213 1238.3
## - Distraction      1      10066 377324 1238.5
## - Class            3      40028 407286 1238.7
## - Game_Frequency   1      10895 378153 1238.8
## - WiFi             1      12477 379736 1239.3
## <none>              367259 1239.9
## - UsePrimaryHand   1      15475 382733 1240.4
## - RefreshRate      2      36741 404000 1242.5
## - CaffeinIntake     1      22391 389650 1242.7
## - AvgSleepTime     1      26198 393457 1243.9
## - InputDevice       4      74330 441589 1244.2
## + Cautious         4        6186 361073 1257.2
##
## Step:  AIC=1229.2
## avg_time ~ Class + Age + AvgSleepTime + LastNightSleep + HoursAweak +
##      Distraction + Gamer + CaffeinIntake + VisualAcuity + PrimarilyHand +
##      UsePrimaryHand + InputDevice + RefreshRate + Device + WiFi +
##      Fatigue_3 + Stress_3 + Noise_Level + Game_Frequency + Sports_Frequency +
##      Shooting_Levels + System + logAvgHoursExercise
##
##              Df Sum of Sq  RSS    AIC
## - System      2      4184 382651 1220.9
## - VisualAcuity 4      35557 414024 1221.3
## - Noise_Level  2       5619 384085 1221.4
## - Device       3      23281 401748 1222.3
## - Shooting_Levels 2      10787 389254 1223.1
## - PrimarilyHand 2      13084 391551 1223.9
## - logAvgHoursExercise 1         4 378471 1224.3
## - LastNightSleep 1         78 378545 1224.4
## - Age          1         538 379005 1224.5
## - HoursAweak   1         650 379117 1224.6
## - Gamer        1        1425 379892 1224.8
## - Sports_Frequency 1        3391 381858 1225.5
## - Stress_3     2      20992 399459 1226.4
## - Class        3      38229 416696 1227.0
## - Game_Frequency 1        8495 386962 1227.2
## - Distraction  1        8793 387260 1227.3
## - Fatigue_3    2      27884 406351 1228.7
## - WiFi         1      13749 392215 1228.9
## <none>          378467 1229.2
## - UsePrimaryHand 1      19388 397855 1230.8
## - InputDevice   4      68310 446777 1231.2
## - CaffeinIntake 1      22626 401093 1231.8

```

```

## - RefreshRate          2      38413 416880 1232.0
## - AvgSleepTime         1      25529 403996 1232.8
## + Temperature          3      11208 367259 1239.9
## + Cautious              4        6549 371918 1246.4
##
## Step: AIC=1220.9
## avg_time ~ Class + Age + AvgSleepTime + LastNightSleep + HoursAweak +
##   Distraction + Gamer + CaffeinIntake + VisualAcuity + PrimarilyHand +
##   UsePrimaryHand + InputDevice + RefreshRate + Device + WiFi +
##   Fatigue_3 + Stress_3 + Noise_Level + Game_Frequency + Sports_Frequency +
##   Shooting_Levels + logAvgHoursExercise
##
##              Df Sum of Sq    RSS    AIC
## - Noise_Level      2      4347 386998 1212.6
## - VisualAcuity     4     36345 418996 1213.2
## - Device            3     23904 406555 1214.1
## - Shooting_Levels   2     11863 394514 1215.1
## - PrimarilyHand     2     13695 396346 1215.7
## - logAvgHoursExercise 1         44 382695 1216.0
## - HoursAweak        1         101 382751 1216.1
## - LastNightSleep    1         309 382960 1216.1
## - Age                1        1275 383925 1216.5
## - Gamer              1        1799 384450 1216.6
## - Sports_Frequency   1        4666 387317 1217.6
## - Class              3     36049 418700 1217.9
## - Game_Frequency     1        8180 390831 1218.8
## - Distraction        1        8254 390905 1218.8
## - Stress_3           2     23453 406104 1218.8
## - Fatigue_3          2     25061 407712 1219.4
## <none>                382651 1220.9
## - WiFi              1     15046 397697 1221.0
## - RefreshRate        2     34523 417174 1222.3
## - UsePrimaryHand     1     20006 402657 1222.6
## - AvgSleepTime       1     23174 405825 1223.6
## - CaffeinIntake       1     24173 406824 1223.9
## + System             2         4184 378467 1229.2
## - InputDevice        4     98241 480891 1230.9
## + Temperature        3     11934 370717 1231.4
## + Cautious           4         7997 374654 1237.6
##
## Step: AIC=1212.63
## avg_time ~ Class + Age + AvgSleepTime + LastNightSleep + HoursAweak +
##   Distraction + Gamer + CaffeinIntake + VisualAcuity + PrimarilyHand +
##   UsePrimaryHand + InputDevice + RefreshRate + Device + WiFi +
##   Fatigue_3 + Stress_3 + Game_Frequency + Sports_Frequency +
##   Shooting_Levels + logAvgHoursExercise
##
##              Df Sum of Sq    RSS    AIC
## - VisualAcuity      4     38779 425777 1205.5
## - Shooting_Levels    2     10264 397262 1206.3
## - PrimarilyHand      2     13663 400661 1207.4
## - logAvgHoursExercise 1          1 386999 1207.8
## - HoursAweak         1          73 387071 1207.8
## - LastNightSleep     1         367 387365 1207.9

```

```

## - Device          3      31210 418208 1208.1
## - Gamer           1       1257 388255 1208.2
## - Age             1       1519 388517 1208.3
## - Class           3      33787 420785 1208.8
## - Sports_Frequency 1       5428 392426 1209.6
## - Distraction     1       6008 393006 1209.8
## - Stress_3        2      23074 410072 1210.4
## - Game_Frequency  1       8251 395250 1210.5
## - Fatigue_3       2      23851 410849 1210.6
## - WiFi            1      14667 401666 1212.6
## <none>              386998 1212.6
## - RefreshRate     2      34545 421543 1213.9
## - UsePrimaryHand  1      22667 409665 1215.1
## - AvgSleepTime    1      23015 410013 1215.2
## - CaffeinIntake   1      23215 410213 1215.3
## + Noise_Level     2       4347 382651 1220.9
## + System          2       2913 384085 1221.4
## - InputDevice     4      96745 483743 1222.0
## + Temperature     3       9469 377529 1224.0
## + Cautious        4       9588 377410 1228.8
##
## Step: AIC=1205.51
## avg_time ~ Class + Age + AvgSleepTime + LastNightSleep + HoursAweak +
##   Distraction + Gamer + CaffeinIntake + PrimarilyHand + UsePrimaryHand +
##   InputDevice + RefreshRate + Device + WiFi + Fatigue_3 + Stress_3 +
##   Game_Frequency + Sports_Frequency + Shooting_Levels + logAvgHoursExercise
##
##           Df Sum of Sq   RSS   AIC
## - Class          3      30472 456249 1199.8
## - Shooting_Levels 2      13920 439697 1199.9
## - Stress_3        2      14882 440659 1200.2
## - Gamer           1         5 425782 1200.7
## - logAvgHoursExercise 1        20 425797 1200.7
## - LastNightSleep  1         55 425832 1200.7
## - HoursAweak      1        126 425903 1200.7
## - Age             1         547 426324 1200.8
## - PrimarilyHand   2      16925 442702 1200.8
## - Device          3      36091 461868 1201.4
## - Sports_Frequency 1       4645 430422 1202.0
## - Game_Frequency  1       6796 432573 1202.7
## - WiFi            1       8600 434376 1203.2
## - Distraction     1       8767 434544 1203.3
## - Fatigue_3       2      29208 454985 1204.3
## - CaffeinIntake   1      14298 440074 1204.9
## <none>              425777 1205.5
## - RefreshRate     2      36358 462134 1206.4
## - AvgSleepTime    1      24160 449937 1207.8
## - UsePrimaryHand  1      26149 451926 1208.3
## + VisualAcuity     4      38779 386998 1212.6
## + Noise_Level     2       6781 418996 1213.2
## + System          2       3121 422656 1214.3
## - InputDevice     4     111805 537582 1216.2
## + Temperature     3       5643 420134 1218.4
## + Cautious        4      12690 413087 1221.0

```

```

##
## Step: AIC=1199.85
## avg_time ~ Age + AvgSleepTime + LastNightSleep + HoursAweak +
##   Distraction + Gamer + CaffeinIntake + PrimarilyHand + UsePrimaryHand +
##   InputDevice + RefreshRate + Device + WiFi + Fatigue_3 + Stress_3 +
##   Game_Frequency + Sports_Frequency + Shooting_Levels + logAvgHoursExercise
##
##           Df Sum of Sq    RSS    AIC
## - PrimarilyHand      2      15200 471449 1194.4
## - Device              3      34447 490696 1194.7
## - logAvgHoursExercise 1         179 456428 1195.0
## - LastNightSleep      1         254 456503 1195.1
## - HoursAweak          1         318 456567 1195.1
## - Gamer               1         525 456774 1195.1
## - Sports_Frequency    1        3518 459767 1196.0
## - Shooting_Levels     2       21649 477898 1196.1
## - Game_Frequency      1        4527 460777 1196.3
## - Fatigue_3           2       23691 479940 1196.7
## - CaffeinIntake       1        7922 464172 1197.2
## - Distraction         1        9878 466127 1197.8
## - WiFi                1       10276 466525 1197.9
## - Stress_3            2       31047 487296 1198.6
## - RefreshRate         2       32563 488812 1199.0
## <none>                                456249 1199.8
## - UsePrimaryHand      1       26461 482710 1202.3
## - AvgSleepTime        1       30040 486289 1203.2
## + Class               3       30472 425777 1205.5
## - InputDevice         4      102228 558477 1206.5
## - Age                 1       42831 499080 1206.6
## + Noise_Level         2        4071 452178 1208.4
## + VisualAcuity        4       35464 420785 1208.8
## + System              2        1296 454953 1209.2
## + Temperature         3        5658 450591 1212.8
## + Cautious            4       11104 445145 1216.1
##
## Step: AIC=1194.36
## avg_time ~ Age + AvgSleepTime + LastNightSleep + HoursAweak +
##   Distraction + Gamer + CaffeinIntake + UsePrimaryHand + InputDevice +
##   RefreshRate + Device + WiFi + Fatigue_3 + Stress_3 + Game_Frequency +
##   Sports_Frequency + Shooting_Levels + logAvgHoursExercise
##
##           Df Sum of Sq    RSS    AIC
## - Device              3       24984 496434 1186.4
## - Shooting_Levels     2       17731 489180 1189.4
## - logAvgHoursExercise 1         35 471484 1189.5
## - LastNightSleep      1         138 471587 1189.5
## - Sports_Frequency    1         909 472359 1189.8
## - Gamer               1         921 472370 1189.8
## - HoursAweak          1        3828 475278 1190.5
## - Game_Frequency      1        4999 476449 1190.9
## - CaffeinIntake       1        6754 478203 1191.3
## - Distraction         1        6974 478423 1191.4
## - WiFi                1        9910 481360 1192.2
## - RefreshRate         2       30567 502017 1192.7

```

```

## - Fatigue_3          2      34343 505793 1193.7
## - UsePrimaryHand     1      17381 488830 1194.2
## <none>                4      471449 1194.4
## - Stress_3           2      41305 512754 1195.5
## - AvgSleepTime       1      28267 499716 1197.0
## + PrimarilyHand      2      15200 456249 1199.8
## - Age                1      43111 514560 1200.8
## + Class              3      28747 442702 1200.8
## - InputDevice        4     110627 582077 1202.1
## + Noise_Level        2        5782 465667 1202.5
## + VisualAcuity       4      38136 433314 1202.9
## + System             2        2180 469269 1203.5
## + Temperature        3        7900 463549 1206.8
## + Cautious           4      13026 458423 1210.2
##
## Step: AIC=1186.44
## avg_time ~ Age + AvgSleepTime + LastNightSleep + HoursAweak +
##   Distraction + Gamer + CaffeinIntake + UsePrimaryHand + InputDevice +
##   RefreshRate + WiFi + Fatigue_3 + Stress_3 + Game_Frequency +
##   Sports_Frequency + Shooting_Levels + logAvgHoursExercise
##
##              Df Sum of Sq    RSS    AIC
## - logAvgHoursExercise  1         17 496451 1181.6
## - LastNightSleep      1         482 496916 1181.7
## - Sports_Frequency    1         771 497205 1181.8
## - Gamer               1        1229 497663 1181.9
## - HoursAweak          1        2382 498815 1182.2
## - Game_Frequency      1        2441 498874 1182.2
## - Shooting_Levels     2       25184 521618 1183.1
## - CaffeinIntake       1        8974 505407 1183.9
## - Distraction         1       10184 506618 1184.2
## - Fatigue_3           2       30325 526759 1184.4
## - WiFi               1       12878 509311 1184.9
## - UsePrimaryHand      1       15112 511545 1185.5
## <none>                 4       496434 1186.4
## - RefreshRate         2       43484 539917 1187.5
## - Stress_3            2       47521 543954 1188.5
## - AvgSleepTime       1       31235 527669 1189.5
## - Age                1       38244 534678 1191.2
## + Class              3       28012 468421 1193.5
## + Device             3       24984 471449 1194.4
## + PrimarilyHand      2         5737 490696 1194.7
## + Noise_Level        2         5195 491238 1194.8
## + VisualAcuity       4       39298 457135 1195.2
## + System             2         1585 494848 1195.8
## + Temperature        3         6437 489997 1199.3
## - InputDevice        4      141996 638430 1199.5
## + Cautious           4       16082 480351 1201.6
##
## Step: AIC=1181.59
## avg_time ~ Age + AvgSleepTime + LastNightSleep + HoursAweak +
##   Distraction + Gamer + CaffeinIntake + UsePrimaryHand + InputDevice +
##   RefreshRate + WiFi + Fatigue_3 + Stress_3 + Game_Frequency +
##   Sports_Frequency + Shooting_Levels

```

```

##
##          Df Sum of Sq    RSS    AIC
## - LastNightSleep      1      540 496991 1176.9
## - Sports_Frequency     1      958 497408 1177.0
## - Gamer                1     1217 497668 1177.0
## - HoursAweak           1     2420 498871 1177.3
## - Game_Frequency       1     2427 498877 1177.3
## - Shooting_Levels      2    26592 523043 1178.6
## - CaffeinIntake        1     9107 505558 1179.1
## - Distraction          1    10562 507013 1179.4
## - Fatigue_3            2    31573 528023 1179.8
## - WiFi                 1    12865 509316 1180.0
## - UsePrimaryHand       1    15160 511611 1180.6
## <none>                  496451 1181.6
## - RefreshRate          2    44380 540831 1182.9
## - Stress_3             2    47944 544395 1183.8
## - AvgSleepTime         1    31500 527951 1184.7
## + logAvgHoursExercise  1       17 496434 1186.4
## - Age                  1    40034 536485 1186.7
## + Class                3    27886 468565 1188.7
## + Device               3    24967 471484 1189.5
## + PrimarilyHand        2     5594 490856 1189.8
## + Noise_Level          2     5212 491238 1189.9
## + VisualAcuity         4    39221 457230 1190.4
## + System               2     1550 494900 1190.9
## + Temperature          3     6451 489999 1194.5
## - InputDevice           4   142428 638879 1194.7
## + Cautious             4     16031 480419 1196.8
##
## Step:  AIC=1176.87
## avg_time ~ Age + AvgSleepTime + HoursAweak + Distraction + Gamer +
##           CaffeinIntake + UsePrimaryHand + InputDevice + RefreshRate +
##           WiFi + Fatigue_3 + Stress_3 + Game_Frequency + Sports_Frequency +
##           Shooting_Levels
##
##          Df Sum of Sq    RSS    AIC
## - Sports_Frequency     1      917 497908 1172.2
## - Gamer                1     1223 498214 1172.3
## - Game_Frequency       1     2348 499339 1172.6
## - HoursAweak           1     2452 499443 1172.6
## - Shooting_Levels      2    26600 523591 1173.9
## - CaffeinIntake        1     9098 506089 1174.3
## - Distraction          1    10276 507267 1174.7
## - WiFi                 1    12476 509467 1175.2
## - Fatigue_3            2    34112 531103 1175.7
## - UsePrimaryHand       1    15027 512017 1175.8
## <none>                  496991 1176.9
## - RefreshRate          2    44203 541193 1178.1
## - Stress_3             2    47405 544396 1178.9
## + LastNightSleep       1      540 496451 1181.6
## + logAvgHoursExercise  1       75 496916 1181.7
## - AvgSleepTime         1    39943 536934 1182.0
## - Age                  1    41602 538593 1182.4
## + Class                3    28221 468770 1183.9

```

```

## + Device          3      25335 471656 1184.7
## + PrimarilyHand   2       5551 491440 1185.1
## + Noise_Level     2       5306 491685 1185.2
## + VisualAcuity    4      38058 458933 1186.0
## + System          2       1434 495557 1186.2
## + Temperature     3       6038 490953 1189.9
## - InputDevice     4     146159 643150 1190.7
## + Cautious        4      15922 481069 1192.1
##
## Step:  AIC=1172.24
## avg_time ~ Age + AvgSleepTime + HoursAweak + Distraction + Gamer +
##           CaffeinIntake + UsePrimaryHand + InputDevice + RefreshRate +
##           WiFi + Fatigue_3 + Stress_3 + Game_Frequency + Shooting_Levels
##
##           Df Sum of Sq    RSS    AIC
## - Gamer          1      1386 499295 1167.7
## - HoursAweak     1      2721 500630 1168.1
## - Game_Frequency 1      3143 501051 1168.2
## - CaffeinIntake   1      8335 506243 1169.5
## - Shooting_Levels 2     28238 526146 1169.6
## - Distraction     1     10235 508143 1170.0
## - WiFi           1     12099 510007 1170.5
## - Fatigue_3       2     33464 531372 1170.9
## - UsePrimaryHand  1     15333 513241 1171.3
## <none>                        497908 1172.2
## - RefreshRate     2     43934 541842 1173.4
## - Stress_3        2     49014 546922 1174.6
## + Sports_Frequency 1       917 496991 1176.9
## + LastNightSleep  1       500 497408 1177.0
## + logAvgHoursExercise 1      325 497583 1177.0
## - AvgSleepTime    1     39834 537743 1177.3
## - Age            1     42854 540762 1178.0
## + Class          3     27684 470225 1179.4
## + Device          3     25150 472758 1180.1
## + Noise_Level     2      5635 492273 1180.5
## + PrimarilyHand   2      4831 493077 1180.7
## + System          2      1516 496392 1181.6
## + VisualAcuity    4     37485 460423 1181.6
## + Temperature     3      5636 492273 1185.4
## - InputDevice     4     145248 643156 1185.8
## + Cautious        4      16828 481080 1187.2
##
## Step:  AIC=1167.74
## avg_time ~ Age + AvgSleepTime + HoursAweak + Distraction + CaffeinIntake +
##           UsePrimaryHand + InputDevice + RefreshRate + WiFi + Fatigue_3 +
##           Stress_3 + Game_Frequency + Shooting_Levels
##
##           Df Sum of Sq    RSS    AIC
## - HoursAweak     1      2588 501882 1163.5
## - Game_Frequency 1      5180 504475 1164.2
## - CaffeinIntake   1     10350 509645 1165.5
## - Distraction     1     10866 510161 1165.7
## - Shooting_Levels 2     33060 532355 1166.3
## - Fatigue_3       2     33861 533156 1166.5

```



```

## - WiFi 1 14221 513515 1166.5
## - UsePrimaryHand 1 18935 518229 1167.7
## <none> 499295 1167.7
## - RefreshRate 2 43903 543198 1168.9
## - Stress_3 2 51385 550680 1170.7
## + Gamer 1 1386 497908 1172.2
## + Sports_Frequency 1 1081 498214 1172.3
## + LastNightSleep 1 503 498792 1172.5
## + logAvgHoursExercise 1 290 499005 1172.5
## - AvgSleepTime 1 39965 539260 1172.8
## - Age 1 43453 542747 1173.7
## + Class 3 28805 470489 1174.7
## + Device 3 25513 473781 1175.6
## + Noise_Level 2 6074 493221 1175.9
## + PrimarilyHand 2 5074 494220 1176.1
## + VisualAcuity 4 38863 460432 1176.7
## + System 2 1724 497571 1177.0
## + Temperature 3 6500 492795 1180.6
## + Cautious 4 15940 483355 1183.0
## - InputDevice 4 155703 654997 1183.3
##
## Step: AIC=1163.55
## avg_time ~ Age + AvgSleepTime + Distraction + CaffeinIntake +
## UsePrimaryHand + InputDevice + RefreshRate + WiFi + Fatigue_3 +
## Stress_3 + Game_Frequency + Shooting_Levels
##
## Df Sum of Sq RSS AIC
## - Game_Frequency 1 4791 506674 1159.9
## - Distraction 1 9976 511858 1161.2
## - CaffeinIntake 1 10020 511902 1161.2
## - Fatigue_3 2 31583 533465 1161.7
## - WiFi 1 13625 515507 1162.2
## - Shooting_Levels 2 33587 535469 1162.2
## <none> 501882 1163.5
## - UsePrimaryHand 1 20303 522185 1163.8
## - RefreshRate 2 41954 543837 1164.2
## - Stress_3 2 49054 550936 1165.9
## + HoursAweak 1 2588 499295 1167.7
## + Sports_Frequency 1 1354 500528 1168.1
## + Gamer 1 1253 500630 1168.1
## - AvgSleepTime 1 38386 540268 1168.2
## + LastNightSleep 1 528 501354 1168.3
## + logAvgHoursExercise 1 469 501413 1168.3
## - Age 1 43493 545375 1169.4
## + Class 3 29478 472404 1170.3
## + PrimarilyHand 2 6881 495002 1171.5
## + Noise_Level 2 5836 496046 1171.8
## + Device 3 23758 478124 1171.9
## + VisualAcuity 4 38846 463036 1172.6
## + System 2 2226 499657 1172.7
## + Temperature 3 6485 495397 1176.5
## + Cautious 4 16785 485098 1178.6
## - InputDevice 4 156592 658474 1179.1
##

```

```
## Step: AIC=1159.92
## avg_time ~ Age + AvgSleepTime + Distraction + CaffeinIntake +
## UsePrimaryHand + InputDevice + RefreshRate + WiFi + Fatigue_3 +
## Stress_3 + Shooting_Levels
```

	Df	Sum of Sq	RSS	AIC
## - Distraction	1	9399	516073	1157.4
## - Fatigue_3	2	30803	537476	1157.8
## - CaffeinIntake	1	11427	518101	1157.9
## - WiFi	1	17303	523976	1159.4
## <none>			506674	1159.9
## - Shooting_Levels	2	40965	547638	1160.2
## - UsePrimaryHand	1	21635	528308	1160.5
## - RefreshRate	2	42254	548928	1160.5
## - Stress_3	2	52770	559444	1163.0
## + Game_Frequency	1	4791	501882	1163.5
## + Gamer	1	3135	503539	1164.0
## + Sports_Frequency	1	2730	503944	1164.1
## - AvgSleepTime	1	36788	543462	1164.1
## + HoursAweak	1	2199	504475	1164.2
## + logAvgHoursExercise	1	480	506193	1164.7
## + LastNightSleep	1	393	506281	1164.7
## - Age	1	47627	554301	1166.7
## + Class	3	28130	478544	1167.1
## + PrimarilyHand	2	7216	499457	1167.8
## + Noise_Level	2	5861	500812	1168.1
## + System	2	3106	503568	1168.8
## + VisualAcuity	4	39132	467542	1169.0
## + Device	3	20460	486214	1169.2
## + Temperature	3	5374	501300	1173.1
## + Cautious	4	15584	491090	1175.3
## - InputDevice	4	161112	667786	1176.1

```
## Step: AIC=1157.43
## avg_time ~ Age + AvgSleepTime + CaffeinIntake + UsePrimaryHand +
## InputDevice + RefreshRate + WiFi + Fatigue_3 + Stress_3 +
## Shooting_Levels
```

	Df	Sum of Sq	RSS	AIC
## - Fatigue_3	2	32434	548507	1155.6
## - WiFi	1	16084	532157	1156.5
## - CaffeinIntake	1	18303	534376	1157.1
## <none>			516073	1157.4
## - Shooting_Levels	2	40953	557026	1157.6
## - RefreshRate	2	44636	560709	1158.4
## - UsePrimaryHand	1	25136	541209	1158.7
## + Distraction	1	9399	506674	1159.9
## - AvgSleepTime	1	34735	550808	1161.0
## + Game_Frequency	1	4215	511858	1161.2
## + Gamer	1	3809	512264	1161.3
## + Sports_Frequency	1	2540	513533	1161.7
## + HoursAweak	1	1426	514647	1161.9
## + logAvgHoursExercise	1	1099	514974	1162.0
## + LastNightSleep	1	167	515906	1162.2

```

## - Stress_3          2      62638 578711 1162.5
## - Age               1      45851 561924 1163.5
## + Class             3      29900 486173 1164.3
## + VisualAcuity      4      43461 472612 1165.5
## + PrimarilyHand     2       5354 510719 1165.8
## + Device            3      23544 492529 1166.0
## + System            2       4423 511650 1166.0
## + Noise_Level       2       3899 512174 1166.2
## + Temperature       3       5094 510979 1170.7
## + Cautious          4      17788 498285 1172.3
## - InputDevice       4     170658 686731 1174.8
##
## Step:  AIC=1155.57
## avg_time ~ Age + AvgSleepTime + CaffeinIntake + UsePrimaryHand +
##      InputDevice + RefreshRate + WiFi + Stress_3 + Shooting_Levels
##
##              Df Sum of Sq    RSS    AIC
## - CaffeinIntake    1      14370 562877 1154.0
## - WiFi             1      18603 567110 1155.0
## <none>                        548507 1155.6
## - RefreshRate      2      49010 597517 1156.9
## - Stress_3         2      50136 598643 1157.1
## + Fatigue_3        2      32434 516073 1157.4
## + Distraction       1      11031 537476 1157.8
## - UsePrimaryHand   1      33482 581989 1158.3
## - Shooting_Levels  2      57337 605844 1158.7
## + Gamer            1       4553 543954 1159.4
## + Game_Frequency   1       3318 545189 1159.7
## + LastNightSleep   1       2410 546097 1159.9
## - AvgSleepTime     1      41446 589953 1160.1
## + HoursAwake       1        952 547555 1160.2
## + Sports_Frequency 1        875 547632 1160.2
## - Age              1      42449 590956 1160.3
## + logAvgHoursExercise 1       194 548313 1160.4
## + PrimarilyHand    2       7688 540819 1163.5
## + VisualAcuity     4      46810 501697 1163.5
## + Class            3      25855 522652 1163.9
## + System           2       4806 543701 1164.2
## + Device           3      23110 525397 1164.6
## + Noise_Level      2       1086 547421 1165.0
## + Temperature      3       8404 540103 1168.2
## + Cautious         4      23783 524724 1169.3
## - InputDevice      4     164875 713382 1170.0
##
## Step:  AIC=1154.05
## avg_time ~ Age + AvgSleepTime + UsePrimaryHand + InputDevice +
##      RefreshRate + WiFi + Stress_3 + Shooting_Levels
##
##              Df Sum of Sq    RSS    AIC
## - WiFi             1      16704 579581 1153.0
## <none>                        562877 1154.0
## - RefreshRate      2      47383 610260 1154.8
## + Distraction       1      17472 545405 1154.8
## - UsePrimaryHand    1      28044 590921 1155.5

```

```

## + CaffeinIntake      1      14370 548507 1155.6
## - Stress_3           2      54950 617826 1156.3
## - Shooting_Levels    2      56682 619558 1156.7
## + Gamer              1       8444 554433 1157.0
## + Fatigue_3          2      28501 534376 1157.1
## + Game_Frequency     1       4306 558570 1157.9
## - AvgSleepTime       1      42193 605070 1158.5
## + LastNightSleep     1       1688 561188 1158.5
## + logAvgHoursExercise 1       1178 561698 1158.6
## + HoursAweak         1       909 561968 1158.7
## + Sports_Frequency   1        96 562781 1158.9
## - Age                1     57490 620367 1161.7
## + System              2       6116 556760 1162.4
## + Device              3     26692 536185 1162.4
## + PrimarilyHand      2       6096 556781 1162.4
## + Noise_Level        2       1298 561579 1163.5
## + Class              3     19876 543000 1164.0
## + VisualAcuity       4     39616 523260 1164.1
## + Temperature        3     10323 552554 1166.2
## + Cautious           4     17666 545210 1169.4
## - InputDevice        4    186676 749553 1171.6
##
## Step:  AIC=1152.96
## avg_time ~ Age + AvgSleepTime + UsePrimaryHand + InputDevice +
##      RefreshRate + Stress_3 + Shooting_Levels
##
##              Df Sum of Sq    RSS    AIC
## - RefreshRate      2      38588 618169 1151.5
## <none>              579581 1153.0
## - Stress_3         2      46025 625606 1153.1
## - UsePrimaryHand    1      25145 604725 1153.6
## - Shooting_Levels   2      50150 629731 1154.0
## + WiFi             1      16704 562877 1154.0
## + Distraction       1      15752 563829 1154.3
## + Gamer            1      12747 566834 1155.0
## + CaffeinIntake     1      12471 567110 1155.0
## + Fatigue_3        2      31821 547760 1155.4
## + Game_Frequency    1       7149 572432 1156.2
## - AvgSleepTime     1     39927 619508 1156.7
## + logAvgHoursExercise 1      1643 577938 1157.5
## + HoursAweak       1      1097 578484 1157.6
## + LastNightSleep    1       552 579029 1157.7
## + Sports_Frequency  1        65 579516 1157.8
## - Age              1     56084 635665 1160.0
## + System           2       8240 571341 1160.8
## + Device           3     28221 551360 1161.1
## + PrimarilyHand    2       5849 573732 1161.4
## + Noise_Level      2       1237 578344 1162.4
## + Class            3     20137 559444 1163.0
## + VisualAcuity     4     37486 542095 1163.8
## + Temperature      3     12005 567576 1164.8
## + Cautious         4     17982 561599 1168.3
## - InputDevice      4    206012 785593 1172.8
##

```

```

## Step: AIC=1151.55
## avg_time ~ Age + AvgSleepTime + UsePrimaryHand + InputDevice +
## Stress_3 + Shooting_Levels
##
##          Df Sum of Sq    RSS    AIC
## - Stress_3      2      31408 649578 1148.2
## - UsePrimaryHand 1      20059 638228 1150.8
## <none>                      618169 1151.5
## - AvgSleepTime    1      28131 646300 1152.4
## + Distraction      1      18466 599703 1152.5
## + RefreshRate      2      38588 579581 1153.0
## + Gamer            1      12207 605962 1153.8
## + CaffeinIntake    1      11635 606535 1154.0
## + Fatigue_3        2      32187 585983 1154.4
## + WiFi             1       7909 610260 1154.8
## + Game_Frequency   1       6661 611508 1155.0
## + logAvgHoursExercise 1       5245 612925 1155.3
## + HoursAweak       1       3800 614370 1155.6
## - Shooting_Levels  2      70043 688212 1155.7
## + LastNightSleep   1        803 617367 1156.2
## + Sports_Frequency 1         16 618153 1156.4
## + Device           3      41828 576341 1157.1
## - Age              1      58968 677138 1158.5
## + System           2       9841 608328 1159.2
## + PrimarilyHand     2       2122 616047 1160.8
## + Noise_Level       2        333 617836 1161.2
## + Class             3      17642 600527 1162.4
## + VisualAcuity      4      38691 579479 1162.7
## + Temperature       3      16185 601985 1162.7
## + Cautious          4      12811 605358 1168.3
## - InputDevice       4     271562 889731 1179.1
##
## Step: AIC=1148.23
## avg_time ~ Age + AvgSleepTime + UsePrimaryHand + InputDevice +
## Shooting_Levels
##
##          Df Sum of Sq    RSS    AIC
## - UsePrimaryHand    1      11011 660588 1145.5
## + Distraction        1      28950 620627 1147.2
## - AvgSleepTime       1      24124 673701 1148.1
## <none>                      649578 1148.2
## + Gamer              1      16505 633072 1149.8
## + CaffeinIntake       1      15521 634056 1150.0
## + Game_Frequency      1       8352 641225 1151.4
## + Stress_3           2      31408 618169 1151.5
## + HoursAweak         1       4977 644601 1152.1
## + WiFi               1       3966 645611 1152.3
## + logAvgHoursExercise 1       1322 648256 1152.8
## + Sports_Frequency    1        647 648930 1153.0
## + LastNightSleep      1         20 649558 1153.1
## + RefreshRate         2      23972 625606 1153.1
## + Device             3      45013 604564 1153.5
## + Fatigue_3          2      21485 628092 1153.6
## - Shooting_Levels     2      81570 731148 1153.8

```

```

## + System                2      11376 638202 1155.7
## + Class                  3      33514 616064 1156.0
## + PrimarilyHand         2       3633 645945 1157.2
## - Age                    1      75656 725234 1157.6
## + Noise_Level           2       1087 648490 1157.7
## + Temperature           3      12004 637574 1160.4
## + VisualAcuity          4      28589 620989 1161.9
## + Cautious              4      18355 631223 1164.0
## - InputDevice           4     261468 911045 1172.4
##
## Step:  AIC=1145.54
## avg_time ~ Age + AvgSleepTime + InputDevice + Shooting_Levels
##
##              Df Sum of Sq    RSS    AIC
## + Distraction      1      29814 630774 1144.4
## - AvgSleepTime     1      21999 682587 1144.9
## <none>                660588 1145.5
## + Gamer            1      21720 638869 1146.1
## + CaffeinIntake     1      11642 648946 1148.1
## + UsePrimaryHand    1      11011 649578 1148.2
## + Game_Frequency    1       8848 651740 1148.7
## - Shooting_Levels   2      73292 733880 1149.4
## + HoursAweak        1       4211 656378 1149.6
## + WiFi              1       3701 656887 1149.7
## + logAvgHoursExercise 1       1219 659369 1150.2
## + Sports_Frequency   1        925 659663 1150.2
## + Fatigue_3          2      25158 635430 1150.2
## + LastNightSleep     1         2 660586 1150.4
## + RefreshRate        2      22553 638035 1150.8
## + Stress_3           2      22360 638228 1150.8
## + Device             3      38732 621856 1152.3
## + Class              3      35674 624914 1153.0
## + System             2      10872 649716 1153.1
## + PrimarilyHand      2       2970 657618 1154.7
## + Noise_Level        2        307 660281 1155.2
## - Age                1      80964 741552 1155.6
## + Temperature        3      13189 647399 1157.5
## + VisualAcuity       4      31287 629301 1158.7
## + Cautious           4      19194 641394 1161.2
## - InputDevice        4     276674 937262 1171.2
##
## Step:  AIC=1144.44
## avg_time ~ Age + AvgSleepTime + InputDevice + Shooting_Levels +
##   Distraction
##
##              Df Sum of Sq    RSS    AIC
## <none>                630774 1144.4
## - AvgSleepTime       1      26881 657655 1145.0
## - Distraction         1      29814 660588 1145.5
## + Gamer              1      16241 614533 1145.9
## + UsePrimaryHand     1      10147 620627 1147.2
## + Game_Frequency     1      10002 620772 1147.2
## - Shooting_Levels    2      66798 697572 1147.7
## + WiFi               1       5756 625018 1148.1

```

```
## + CaffeinIntake      1      4188 626586 1148.4
## + logAvgHoursExercise 1      2644 628130 1148.8
## + HoursAweak         1      2039 628735 1148.9
## + Fatigue_3          2      24828 605946 1149.0
## + Sports_Frequency   1      1474 629300 1149.0
## + LastNightSleep     1       333 630441 1149.2
## + RefreshRate        2      21687 609087 1149.7
## + Stress_3           2      14083 616691 1151.2
## + Device             3      31374 599400 1152.4
## + Class              3      30755 600019 1152.6
## + Noise_Level        2       6423 624351 1152.8
## + System             2       6359 624415 1152.8
## + PrimarilyHand      2       3173 627601 1153.5
## - Age                1      75201 705976 1154.1
## + Temperature        3      14039 616735 1156.1
## + VisualAcuity       4      27428 603346 1158.1
## + Cautious           4      14802 615972 1160.8
## - InputDevice        4      247894 878668 1167.8
```

```
##
```

```
## Call:
```

```
## lm(formula = avg_time ~ Age + AvgSleepTime + InputDevice + Shooting_Levels +
##     Distraction, data = Survey_cleaned)
```

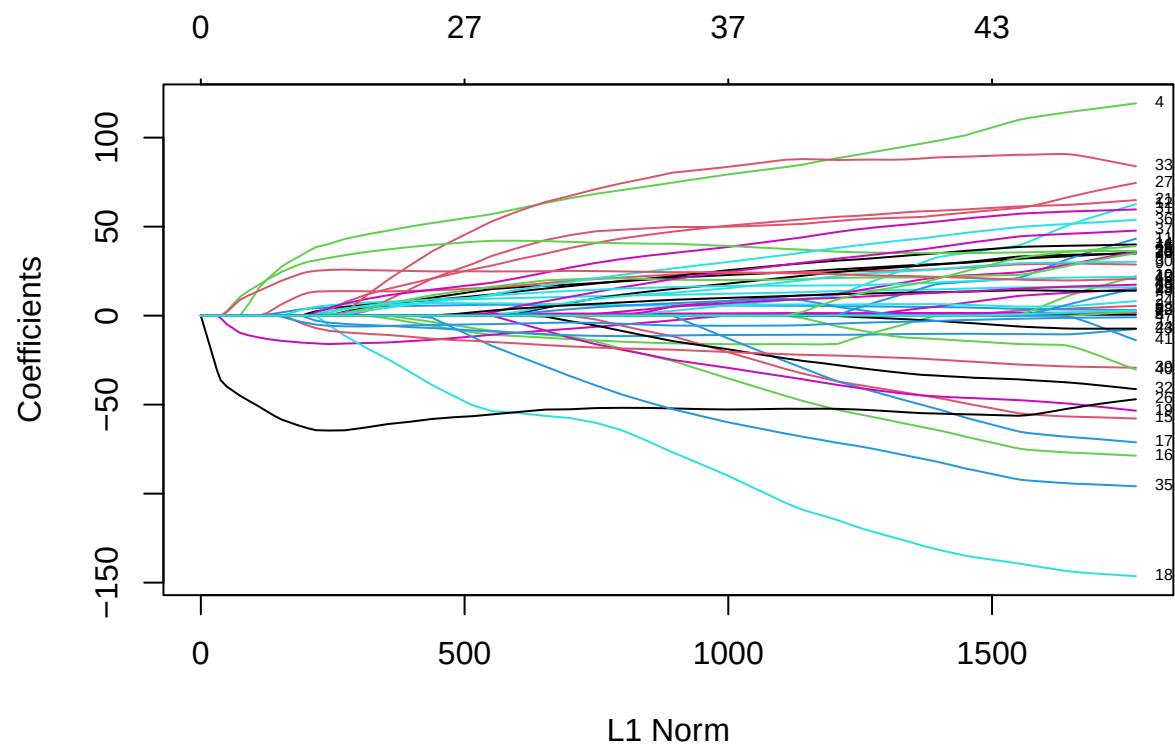
```
##
```

```
## Coefficients:
```

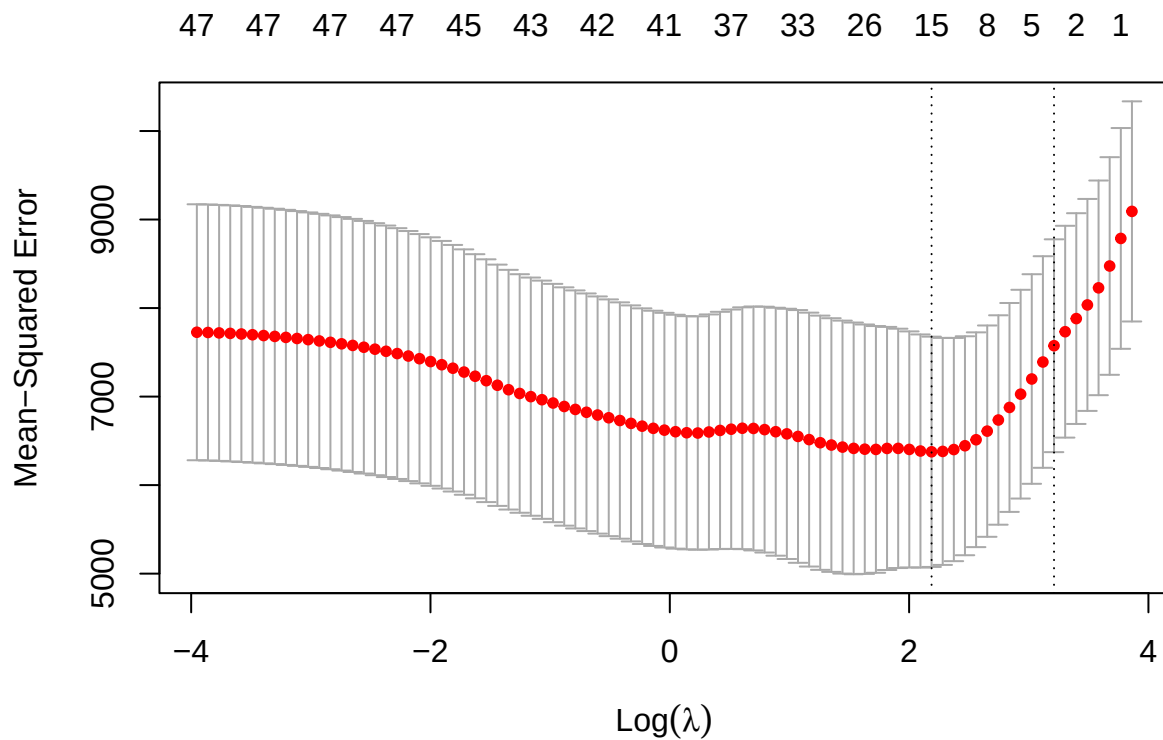
```
##              (Intercept)                      Age
##              113.638                      15.768
##              AvgSleepTime                InputDeviceMouse
##              14.512                      -88.056
##              InputDeviceKeyboard          InputDeviceTrackpad
##              56.524                      8.402
##              InputDeviceTouch screen  Shooting_LevelsIntermediate Levels
##              -28.410                      22.652
##              Shooting_LevelsLow Levels                DistractionYes
##              58.928                      37.854
```

```
## Lasso
```

```
modlasso <- glmnet(model.matrix(avg_time ~ . -avg_time, data = Survey_cleaned),
                    Survey_cleaned$avg_time, alpha = 1)
plot(modlasso, label = TRUE)
```



```
cvmodlasso <- cv.glmnet(model.matrix(avg_time ~ . -avg_time, data = Survey_cleaned),
                        Survey_cleaned$avg_time, alpha = 1)
plot(cvmodlasso)
```

```
bestlamlasso <- cvmodlasso$lambda.min
predict(modlasso, s = bestlamlasso, type = "coefficients")
```

```
## 49 x 1 sparse Matrix of class "dgCMatrix"
##
## (Intercept)          s1
## (Intercept)          .
## ClassJunior          .
## ClassSenior          .
## ClassGraduate student 38.495659
## Age                  4.207794
## AvgSleepTime         4.959268
## LastNightSleep       .
## HoursAweak           .
## DistractionYes       13.502323
## TemperatureWarm      .
## TemperatureNeutral   .
## TemperatureCold      .
## GamerYes             -15.617365
## CaffeinIntakeYes      2.454916
## VisualAcuityExcellent .
## VisualAcuityGood     .
## VisualAcuityPoor     .
## VisualAcuityVery Poor .
## PrimarilyHandLeft hand .
```

```

## PrimarilyHandAmbidextrous (both hands equally) .
## UsePrimaryHandNo .
## CautiousVery cautious .
## CautiousModerately cautious -2.849909
## CautiousSlightly cautious 1.386369
## CautiousNot cautious at all .
## InputDeviceMouse -64.358406
## InputDeviceKeyboard .
## InputDeviceTrackpad 0.167877
## InputDeviceTouch screen .
## RefreshRateMid-Range .
## RefreshRateStandard 2.010429
## DeviceiPad(Tablet .
## DeviceiPhone(Smartphone .
## DeviceLaptop 31.126876
## WiFiStable .
## Fatigue_3Moderately Fatigue .
## Fatigue_3Low Fatigue .
## Stress_3Moderately Stress .
## Stress_3Low Stress -7.020384
## Noise_LevelModerately Noise .
## Noise_LevelLow Noise .
## Game_FrequencyLow Frequency .
## Sports_FrequencyLow Frequency .
## Shooting_LevelsIntermediate Levels .
## Shooting_LevelsLow Levels 25.202538
## SystemLinux .
## SystemWindows -5.059475
## logAvgHoursExercise .

```

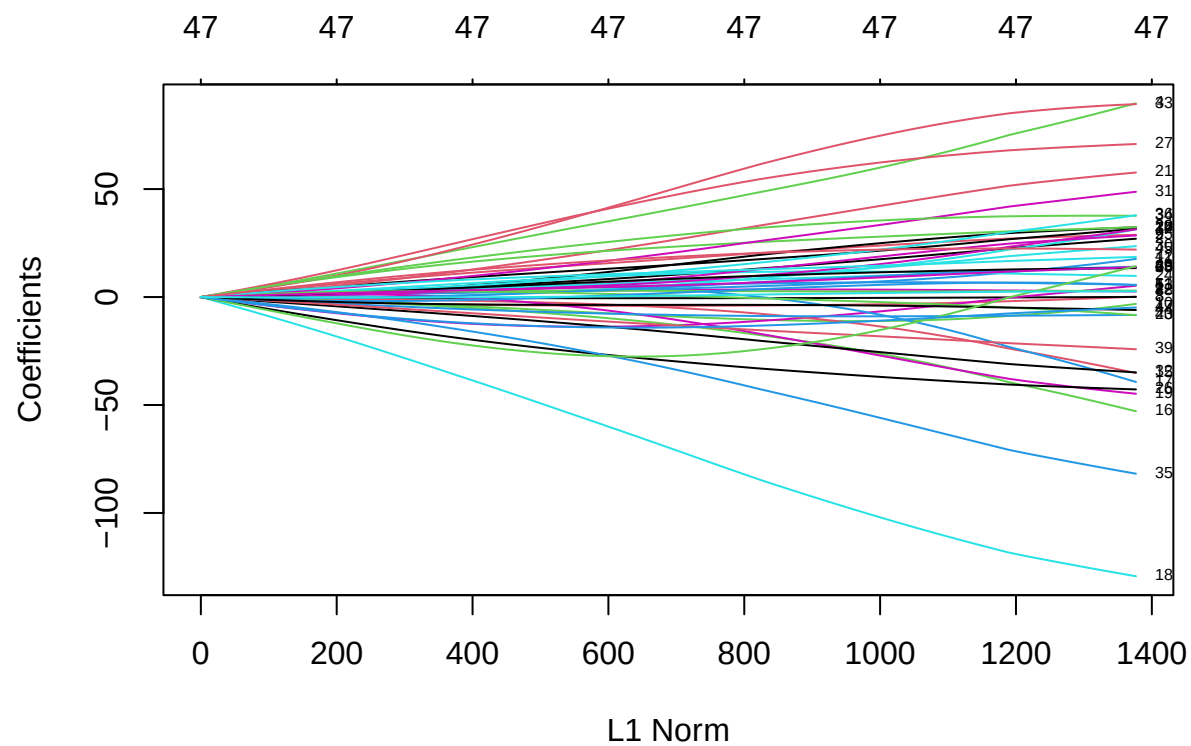
```
lasso.coef <- round(predict(modlasso, s=bestlamlasso, type = "coefficients")[1:49,],2)
```

```
## Ridge
```

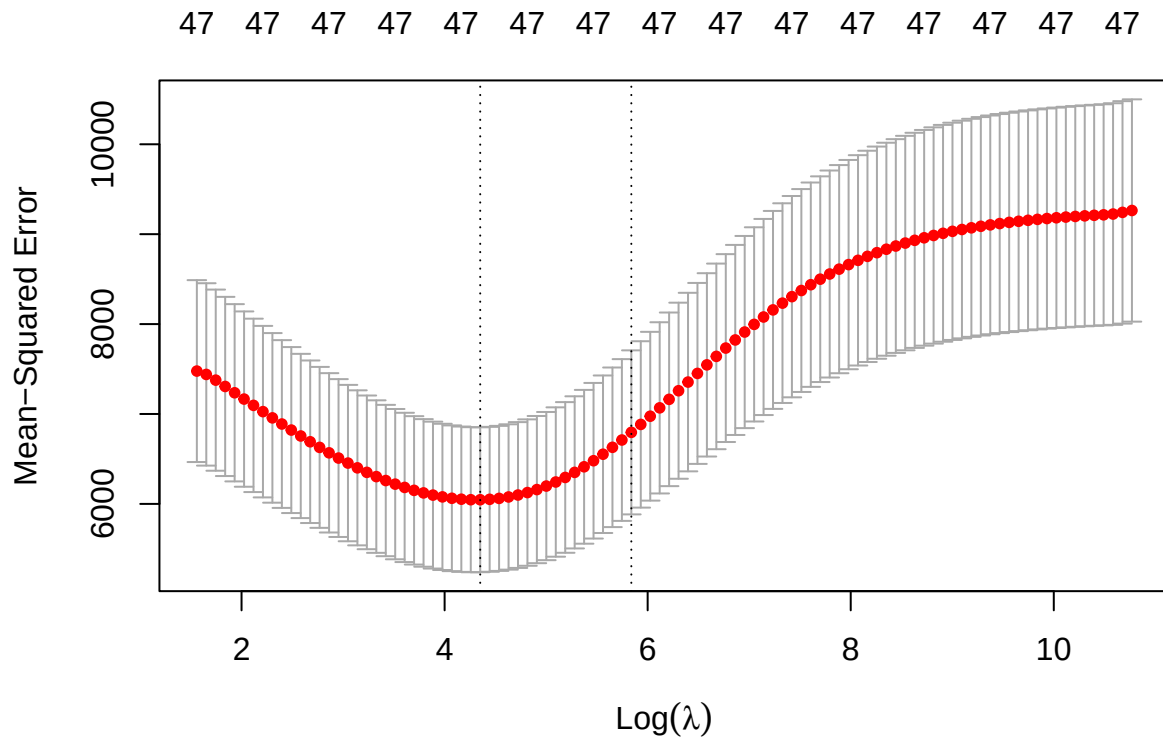
```

modridge <- glmnet(model.matrix(avg_time ~ . -avg_time, data = Survey_cleaned),
                    Survey_cleaned$avg_time, alpha = 0)
plot(modridge, label = TRUE)

```



```
cvmodridge <- cv.glmnet(model.matrix(avg_time ~ . -avg_time, data = Survey_cleaned),
                        Survey_cleaned$avg_time, alpha = 0)
plot(cvmodridge)
```



```
bestlamridge <- cvmodridge$lambda.min
predict(modridge, s = bestlamridge, type = "coefficients")
```

```
## 49 x 1 sparse Matrix of class "dgCMatrix"
##
## (Intercept)          s1
## (Intercept)          .
## ClassJunior          9.2179213
## ClassSenior         -4.0525003
## ClassGraduate student 37.7032688
## Age                  5.1690070
## AvgSleepTime         6.5175895
## LastNightSleep       3.1701663
## HoursAweak          -0.5404607
## DistractionYes       16.6195321
## TemperatureWarm     -8.2325000
## TemperatureNeutral   3.5619055
## TemperatureCold      4.6298805
## GamerYes            -13.5314439
## CaffeinIntakeYes     13.9420796
## VisualAcuityExcellent -4.0959600
## VisualAcuityGood     -11.0979381
## VisualAcuityPoor      4.1943588
## VisualAcuityVery Poor -64.8507779
## PrimarilyHandLeft hand -8.0711916
```

```
## PrimarilyHandAmbidextrous (both hands equally) 13.1496551
## UsePrimaryHandNo 23.8720903
## CautiousVery cautious 2.9004930
## CautiousModerately cautious -7.9892714
## CautiousSlightly cautious 10.2090207
## CautiousNot cautious at all 7.5536393
## InputDeviceMouse -28.1947413
## InputDeviceKeyboard 43.8645305
## InputDeviceTrackpad 22.3201868
## InputDeviceTouch screen 3.9222785
## RefreshRateMid-Range 6.6727554
## RefreshRateStandard 18.5185918
## DeviceiPad(Tablet -14.9492712
## DeviceiPhone(Smartphone 45.4042015
## DeviceLaptop 26.9626098
## WiFiStable -29.9684480
## Fatigue_3Moderately Fatigue 10.8075295
## Fatigue_3Low Fatigue 6.1120911
## Stress_3Moderately Stress 7.7363361
## Stress_3Low Stress -12.1926836
## Noise_LevelModerately Noise 1.0909481
## Noise_LevelLow Noise 1.2814921
## Game_FrequencyLow Frequency 11.0270908
## Sports_FrequencyLow Frequency 4.9663124
## Shooting_LevelsIntermediate Levels -3.6278837
## Shooting_LevelsLow Levels 17.7413953
## SystemLinux -27.5803947
## SystemWindows -14.1677818
## logAvgHoursExercise 0.4618736
```

```
ridge.coef <- round(predict(modridge, s=bestlamridge, type = "coefficients")[1:49,],2)
```

```
# Model with AIC
AICmod <- lm(avg_time ~ Class + AvgSleepTime + CaffeinIntake + VisualAcuity +
             PrimarilyHand + UsePrimaryHand + InputDevice + RefreshRate + Device + WiFi +
             Fatigue_3 + Stress_3 + Game_Frequency, data = Survey_cleaned)
summary(AICmod)
```

```
##
## Call:
## lm(formula = avg_time ~ Class + AvgSleepTime + CaffeinIntake +
##     VisualAcuity + PrimarilyHand + UsePrimaryHand + InputDevice +
##     RefreshRate + Device + WiFi + Fatigue_3 + Stress_3 + Game_Frequency,
##     data = Survey_cleaned)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -138.894  -31.697   -6.822   33.277  192.712
##
## Coefficients:
##                                Estimate Std. Error t value
## (Intercept)                   425.317    105.952    4.014
## ClassJunior                    38.217     15.069    2.536
```

## ClassSenior	13.981	18.142	0.771
## ClassGraduate student	130.035	28.253	4.602
## AvgSleepTime	15.003	6.128	2.448
## CaffeinIntakeYes	39.404	14.849	2.654
## VisualAcuityExcellent	-69.883	25.778	-2.711
## VisualAcuityGood	-82.245	26.164	-3.143
## VisualAcuityPoor	-80.451	42.878	-1.876
## VisualAcuityVery Poor	-117.412	75.133	-1.563
## PrimarilyHandLeft hand	-41.362	35.280	-1.172
## PrimarilyHandAmbidextrous (both hands equally)	32.110	36.761	0.873
## UsePrimaryHandNo	71.433	33.181	2.153
## InputDeviceMouse	-31.564	68.331	-0.462
## InputDeviceKeyboard	68.597	85.193	0.805
## InputDeviceTrackpad	45.508	67.390	0.675
## InputDeviceTouch screen	-3.979	75.914	-0.052
## RefreshRateMid-Range	28.670	19.571	1.465
## RefreshRateStandard	55.821	18.704	2.984
## DeviceiPad(Tablet	-11.942	53.720	-0.222
## DeviceiPhone(Smartphone	140.629	63.237	2.224
## DeviceLaptop	38.479	21.151	1.819
## WiFiStable	-65.223	49.872	-1.308
## Fatigue_3Moderately Fatigue	62.065	20.502	3.027
## Fatigue_3Low Fatigue	54.603	20.072	2.720
## Stress_3Moderately Stress	7.351	17.207	0.427
## Stress_3Low Stress	-35.489	18.291	-1.940
## Game_FrequencyLow Frequency	26.502	12.463	2.127
##	Pr(> t)		
## (Intercept)	0.000115	***	
## ClassJunior	0.012744	*	
## ClassSenior	0.442730		
## ClassGraduate student	1.22e-05	***	
## AvgSleepTime	0.016081	*	
## CaffeinIntakeYes	0.009250	**	
## VisualAcuityExcellent	0.007885	**	
## VisualAcuityGood	0.002193	**	
## VisualAcuityPoor	0.063504	.	
## VisualAcuityVery Poor	0.121248		
## PrimarilyHandLeft hand	0.243804		
## PrimarilyHandAmbidextrous (both hands equally)	0.384468		
## UsePrimaryHandNo	0.033711	*	
## InputDeviceMouse	0.645123		
## InputDeviceKeyboard	0.422595		
## InputDeviceTrackpad	0.501031		
## InputDeviceTouch screen	0.958303		
## RefreshRateMid-Range	0.146044		
## RefreshRateStandard	0.003564	**	
## DeviceiPad(Tablet	0.824536		
## DeviceiPhone(Smartphone	0.028387	*	
## DeviceLaptop	0.071840	.	
## WiFiStable	0.193907		
## Fatigue_3Moderately Fatigue	0.003131	**	
## Fatigue_3Low Fatigue	0.007680	**	
## Stress_3Moderately Stress	0.670124		
## Stress_3Low Stress	0.055140	.	

```
## Game_FrequencyLow Frequency          0.035894 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 64.09 on 101 degrees of freedom
## Multiple R-squared:  0.6454, Adjusted R-squared:  0.5506
## F-statistic: 6.808 on 27 and 101 DF,  p-value: 4.733e-13
```

```
## Collinearity -- VIF only because we include categorical variables
vif(AICmod)
```

```
##              GVIF Df GVIF^(1/(2*Df))
## Class          1.882939 3      1.111235
## AvgSleepTime    1.242119 1      1.114504
## CaffeinIntake    1.291534 1      1.136457
## VisualAcuity     1.918683 4      1.084864
## PrimarilyHand    2.157614 2      1.211974
## UsePrimaryHand   1.774313 1      1.332033
## InputDevice      6.483538 4      1.263213
## RefreshRate      2.266885 2      1.227036
## Device           6.241066 3      1.356885
## WiFi             1.192216 1      1.091886
## Fatigue_3        1.749256 2      1.150041
## Stress_3         1.897225 2      1.173626
## Game_Frequency   1.198209 1      1.094627
```

```
## Remove InputDevice due to the high VIF
```

```
AICmod2 <- lm(avg_time ~ Class + AvgSleepTime + CaffeinIntake + VisualAcuity +
  UsePrimaryHand + RefreshRate + Device + WiFi + Fatigue_3 + Stress_3 +
  Game_Frequency + AvgSleepTime:Class + AvgSleepTime:CaffeinIntake +
  AvgSleepTime:VisualAcuity + AvgSleepTime:PrimarilyHand +
  AvgSleepTime:UsePrimaryHand + AvgSleepTime:RefreshRate + AvgSleepTime:Device +
  AvgSleepTime:WiFi + AvgSleepTime:Fatigue_3 + AvgSleepTime:Stress_3 +
  AvgSleepTime:Game_Frequency,
  data = Survey_cleaned)
anova(AICmod2)
```

```
## Analysis of Variance Table
```

```
##
## Response: avg_time
##              Df Sum Sq Mean Sq F value    Pr(>F)
## Class          3  70764   23588   4.6315 0.0047165 **
## AvgSleepTime    1   51358   51358  10.0842 0.0020692 **
## CaffeinIntake    1   44474   44474   8.7324 0.0040213 **
## VisualAcuity     4  112552   28138   5.5249 0.0005194 ***
## UsePrimaryHand    1    8500    8500   1.6689 0.1998229
## RefreshRate      2  133463   66731  13.1028 1.061e-05 ***
## Device           3   90243   30081   5.9064 0.0010229 **
## WiFi             1   19165   19165   3.7631 0.0556363 .
## Fatigue_3        2   29443   14722   2.8906 0.0608947 .
## Stress_3         2   33627   16814   3.3014 0.0414977 *
## Game_Frequency    1   31181   31181   6.1224 0.0152893 *
## Class:AvgSleepTime 3    4805    1602   0.3145 0.8148506
```

```
## AvgSleepTime:CaffeinIntake    1    2820    2820    0.5538 0.4587899
## AvgSleepTime:VisualAcuity     3    4600    1533    0.3011 0.8245341
## AvgSleepTime:PrimarilyHand    2   19859    9929    1.9497 0.1484902
## AvgSleepTime:UsePrimaryHand   1    4988    4988    0.9794 0.3250906
## AvgSleepTime:RefreshRate      2   21825   10912    2.1426 0.1234963
## AvgSleepTime:Device           2   20285   10143    1.9915 0.1426606
## AvgSleepTime:WiFi             1     162     162    0.0318 0.8588795
## AvgSleepTime:Fatigue_3        2   13297    6648    1.3054 0.2763119
## AvgSleepTime:Stress_3         2    8767    4383    0.8607 0.4264506
## AvgSleepTime:Game_Frequency   1     694     694    0.1363 0.7129253
## Residuals                     87 443084    5093
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
AICmod3 <- lm(avg_time ~ Class + AvgSleepTime + CaffeinIntake + VisualAcuity +
               UsePrimaryHand + RefreshRate + Device + WiFi + Fatigue_3 + Stress_3 +
               Game_Frequency, data = Survey_cleaned)
anova(AICmod3)
```

```
## Analysis of Variance Table
##
## Response: avg_time
##              Df Sum Sq Mean Sq F value    Pr(>F)
## Class         3  70764   23588   4.6294 0.0043877 **
## AvgSleepTime   1  51358   51358  10.0797 0.0019583 **
## CaffeinIntake  1  44474   44474   8.7285 0.0038528 **
## VisualAcuity   4 112552   28138   5.5225 0.0004426 ***
## UsePrimaryHand 1   8500    8500   1.6682 0.1992846
## RefreshRate    2 133463   66731  13.0969 8.168e-06 ***
## Device         3  90243   30081   5.9038 0.0009104 ***
## WiFi          1  19165   19165   3.7614 0.0550824 .
## Fatigue_3      2  29443   14722   2.8893 0.0599655 .
## Stress_3       2  33627   16814   3.2999 0.0406746 *
## Game_Frequency 1  31181   31181   6.1196 0.0149411 *
## Residuals     107 545186    5095
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
AICmod4 <- lm(avg_time ~ Class + AvgSleepTime + CaffeinIntake + VisualAcuity + RefreshRate +
               Device + Fatigue_3 + Game_Frequency + UsePrimaryHand, data = Survey_cleaned)
anova(AICmod4)
```

```
## Analysis of Variance Table
##
## Response: avg_time
##              Df Sum Sq Mean Sq F value    Pr(>F)
## Class         3  70764   23588   4.4860 0.0052020 **
## AvgSleepTime   1  51358   51358   9.7673 0.0022721 **
## CaffeinIntake  1  44474   44474   8.4581 0.0043975 **
## VisualAcuity   4 112552   28138   5.3513 0.0005638 ***
## RefreshRate    2 124040   62020  11.7950 2.286e-05 ***
## Device         3  82181   27394   5.2098 0.0021120 **
## Fatigue_3      2  39237   19619   3.7311 0.0270521 *
```



```
## Game_Frequency    1  44518   44518  8.4664 0.0043787 **
## UsePrimaryHand    1  22436   22436  4.2668 0.0412128 *
## Residuals        110 578395    5258
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Model with BIC
```

```
BICmod <- lm(avg_time ~ Age + AvgSleepTime + InputDevice + Shooting_Levels +
             Distraction, data = Survey_cleaned)
```

```
## Collinearity -- VIF only because we include categorical variables
```

```
vif(BICmod)
```

```
##              GVIF Df GVIF^(1/(2*Df))
## Age              1.083396 1          1.040863
## AvgSleepTime     1.064404 1          1.031699
## InputDevice       1.183969 4          1.021333
## Shooting_Levels  1.095841 2          1.023144
## Distraction       1.053618 1          1.026459
```

```
summary(BICmod)
```

```
##
## Call:
## lm(formula = avg_time ~ Age + AvgSleepTime + InputDevice + Shooting_Levels +
##     Distraction, data = Survey_cleaned)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -128.316  -49.446   -2.732   51.405  201.703
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    113.638    124.733   0.911  0.364110
## Age              15.768     4.186   3.767  0.000259 ***
## AvgSleepTime    14.512     6.444   2.252  0.026159 *
## InputDeviceMouse -88.056    74.274  -1.186  0.238156
## InputDeviceKeyboard  56.524    89.861   0.629  0.530541
## InputDeviceTrackpad  8.402    73.690   0.114  0.909417
## InputDeviceTouch screen -28.410    76.999  -0.369  0.712813
## Shooting_LevelsIntermediate Levels  22.652    20.549   1.102  0.272535
## Shooting_LevelsLow Levels    58.928    18.504   3.185  0.001851 **
## DistractionYes    37.854    15.961   2.372  0.019315 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 72.81 on 119 degrees of freedom
## Multiple R-squared:  0.4609, Adjusted R-squared:  0.4201
## F-statistic: 11.3 on 9 and 119 DF, p-value: 1.231e-12
```

```
## Remove InputDevice due to high VIF
```

```
BICmod2 <- lm(avg_time ~ Age + AvgSleepTime + Shooting_Levels +
```

```

    Distraction + Age:Shooting_Levels + Age:Distraction + AvgSleepTime:Shooting_Levels +
    AvgSleepTime:Distraction, data = Survey_cleaned)
anova(BICmod2)

```

```

## Analysis of Variance Table
##
## Response: avg_time
##
##           Df Sum Sq Mean Sq F value    Pr(>F)
## Age           1   37863    37863   5.9245 0.0164430 *
## AvgSleepTime   1   60340    60340   9.4417 0.0026391 **
## Shooting_Levels 2  134489    67245  10.5220 6.278e-05 ***
## Distraction    1   58594    58594   9.1684 0.0030297 **
## Age:Shooting_Levels 2  120026    60013   9.3905 0.0001651 ***
## Age:Distraction 1    1556     1556   0.2435 0.6225874
## AvgSleepTime:Shooting_Levels 2    9066     4533   0.7093 0.4941076
## AvgSleepTime:Distraction 1     288      288   0.0451 0.8321666
## Residuals    117  747731     6391
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

BICmod3 <- lm(avg_time ~ Age + AvgSleepTime + Shooting_Levels +
    Distraction + Age:Shooting_Levels, data = Survey_cleaned)
anova(BICmod3)

```

```

## Analysis of Variance Table
##
## Response: avg_time
##
##           Df Sum Sq Mean Sq F value    Pr(>F)
## Age           1   37863    37863   6.0390 0.0154107 *
## AvgSleepTime   1   60340    60340   9.6240 0.0023906 **
## Shooting_Levels 2  134489    67245  10.7252 5.151e-05 ***
## Distraction    1   58594    58594   9.3455 0.0027516 **
## Age:Shooting_Levels 2  120026    60013   9.5718 0.0001383 ***
## Residuals    121  758642     6270
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

# Lasso prediction
X_new <- model.matrix(~ . - avg_time, data = Survey_cleaned)
predlasso <- predict(modlasso, newx = X_new, s = bestlamlasso)

```

```

# Ridge prediction
predridge <- predict(modridge, newx = X_new, s = bestlamridge)

```

```

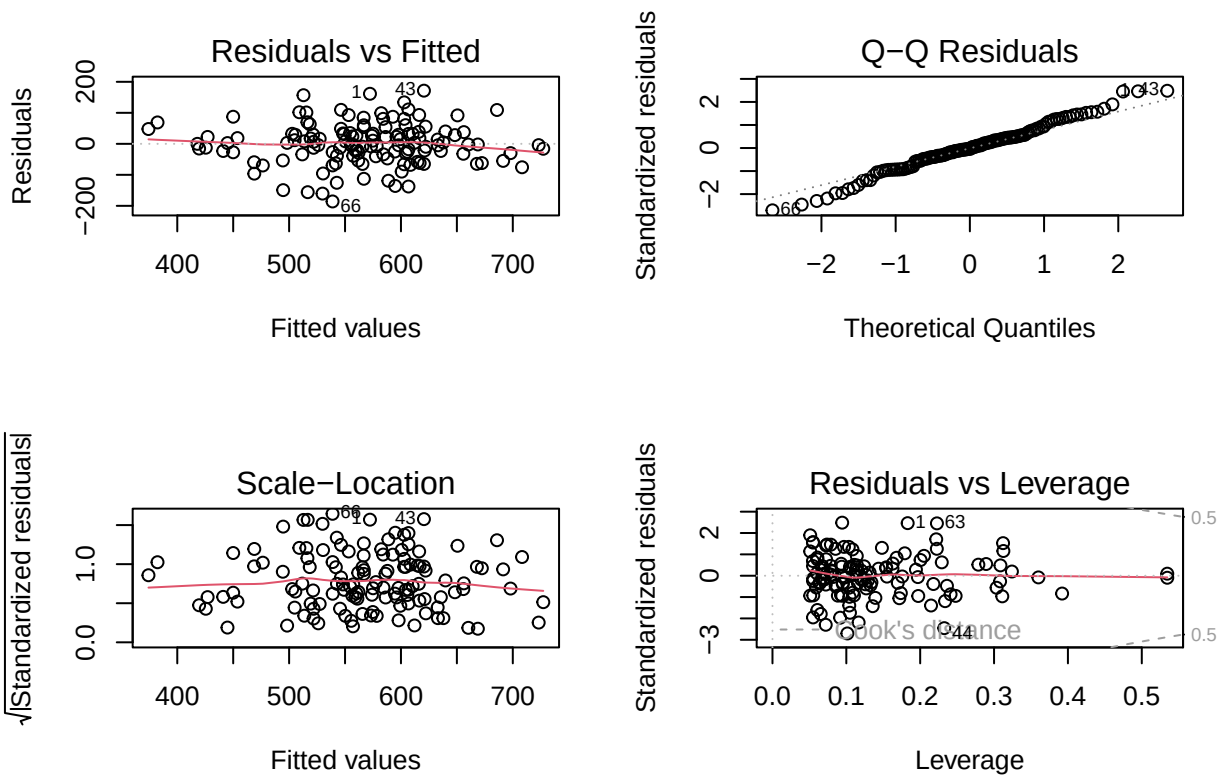
# Model Diagnostics AIC
par(mfrow = c(2,2))
plot(AICmod4)

```

```

## Warning: not plotting observations with leverage one:
## 14

```



```
par(mfrow = c(1,1))
bptest(AICmod4)
```

```
##
## studentized Breusch-Pagan test
##
## data: AICmod4
## BP = 26.426, df = 18, p-value = 0.0904
```

```
shapiro.test(residuals(AICmod4))
```

```
##
## Shapiro-Wilk normality test
##
## data: residuals(AICmod4)
## W = 0.98837, p-value = 0.3476
```

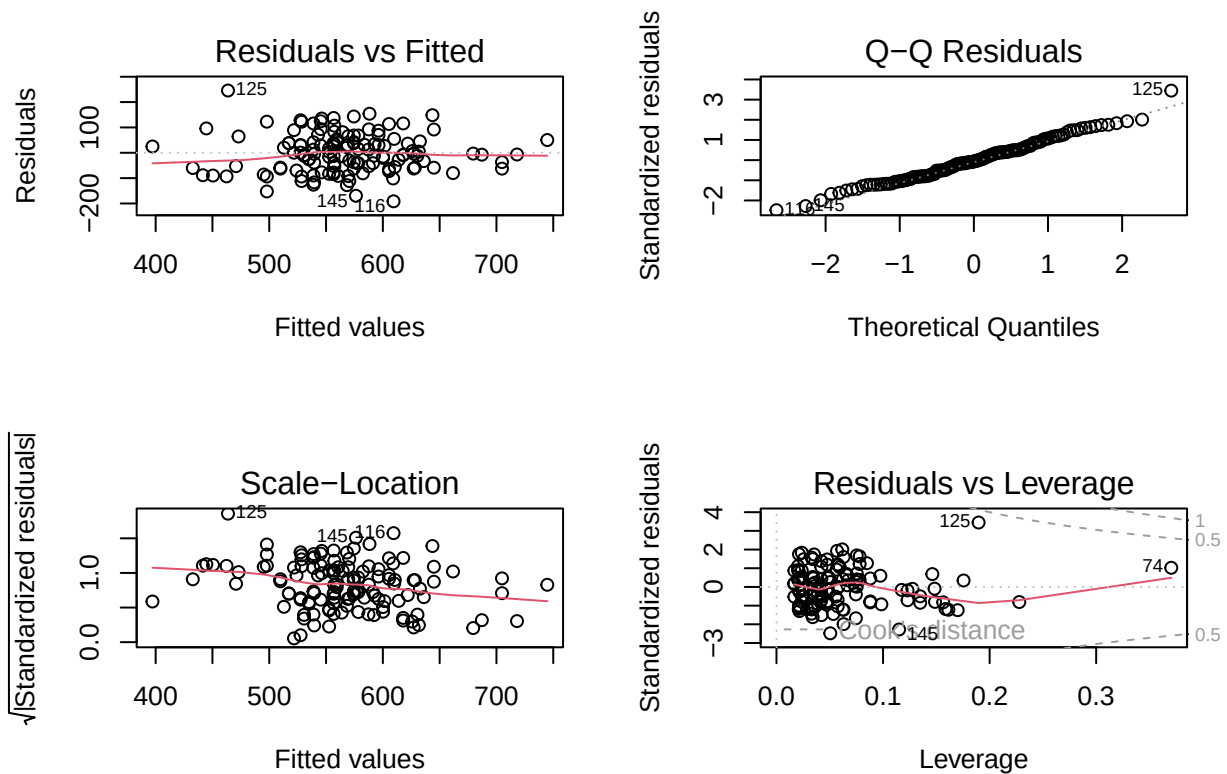
```
dwtest(AICmod4)
```

```
##
## Durbin-Watson test
##
## data: AICmod4
## DW = 2.1674, p-value = 0.8238
## alternative hypothesis: true autocorrelation is greater than 0
```

```
# Model Diagnostics BIC
```

```
par(mfrow = c(2,2))
```

```
plot(BICmod3)
```



```
par(mfrow = c(1,1))
```

```
bptest(BICmod3)
```

```
##
```

```
## studentized Breusch-Pagan test
```

```
##
```

```
## data: BICmod3
```

```
## BP = 18.42, df = 7, p-value = 0.01021
```

```
shapiro.test(residuals(BICmod3))
```

```
##
```

```
## Shapiro-Wilk normality test
```

```
##
```

```
## data: residuals(BICmod3)
```

```
## W = 0.99066, p-value = 0.5411
```

```
dwtest(BICmod3)
```

```

##
## Durbin-Watson test
##
## data: BICmod3
## DW = 1.649, p-value = 0.02156
## alternative hypothesis: true autocorrelation is greater than 0

## WLS -- Non-constant variance Issue
Shooting_Levels_freq <- table(Survey_cleaned$Shooting_Levels) / nrow(Survey_cleaned)
Survey_cleaned$Shooting_Levels_weight <- Shooting_Levels_freq[Survey_cleaned$Shooting_Levels]
BICmod3_weighted <- lm(avg_time ~ Age + AvgSleepTime + Shooting_Levels +
                        Distraction + Age:Shooting_Levels,
                        data = Survey_cleaned, weights = Shooting_Levels_weight)
bptest(BICmod3_weighted)

##
## studentized Breusch-Pagan test
##
## data: BICmod3_weighted
## BP = 9.2754, df = 7, p-value = 0.2335

shapiro.test(residuals(BICmod3_weighted))

##
## Shapiro-Wilk normality test
##
## data: residuals(BICmod3_weighted)
## W = 0.99034, p-value = 0.511

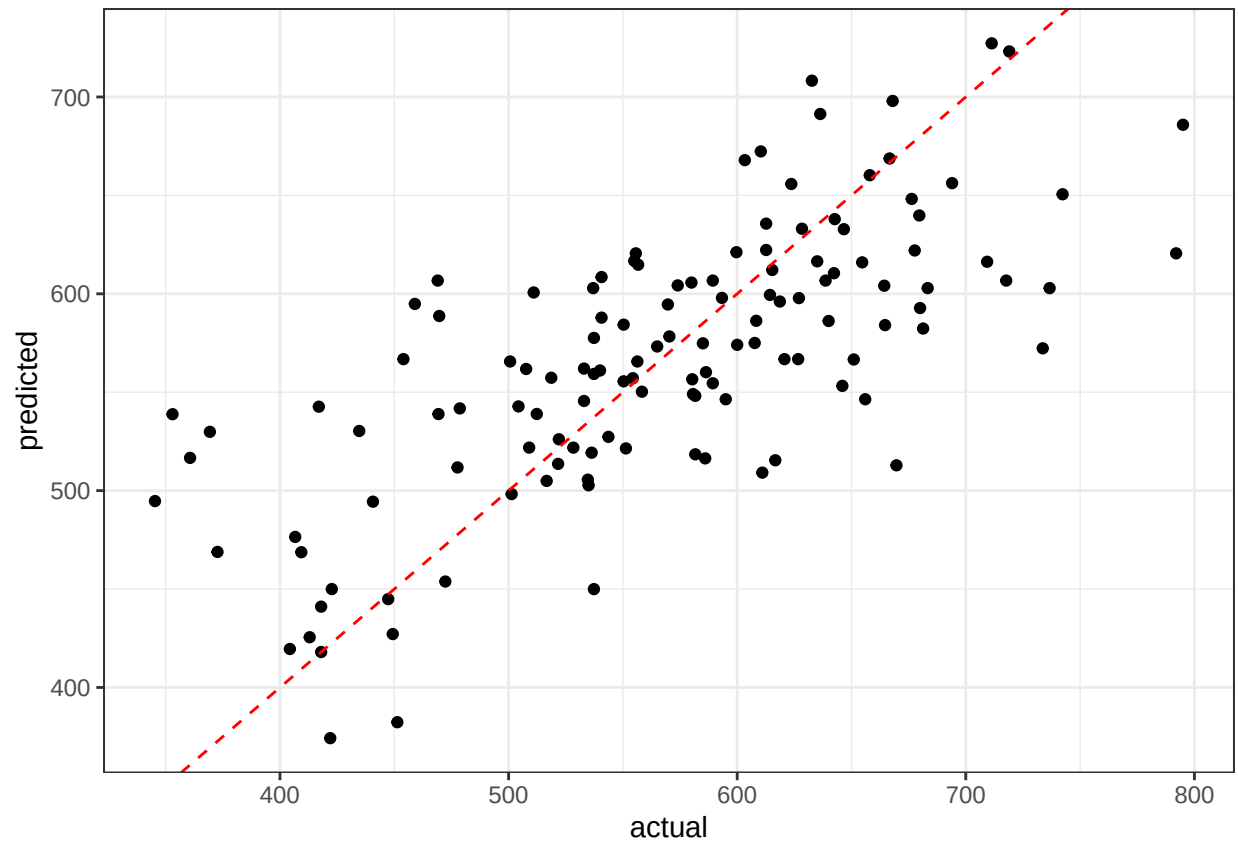
# Check models accuracy
## Define rmse function
rmse <- function(x,y) sqrt(mean((x - y)^2))

## AIC model
rmse(predict(AICmod4), Survey_cleaned$avg_time)

## [1] 66.96031

fitted_AIC <- data.frame(
  actual = Survey_cleaned$avg_time,
  predicted = predict(AICmod4)
)
fitted_AIC %>% ggplot(aes(x = actual, y = predicted)) +
  geom_point() +
  geom_abline(intercept = 0, slope = 1, color = "red", linetype = "dashed") +
  theme_bw()

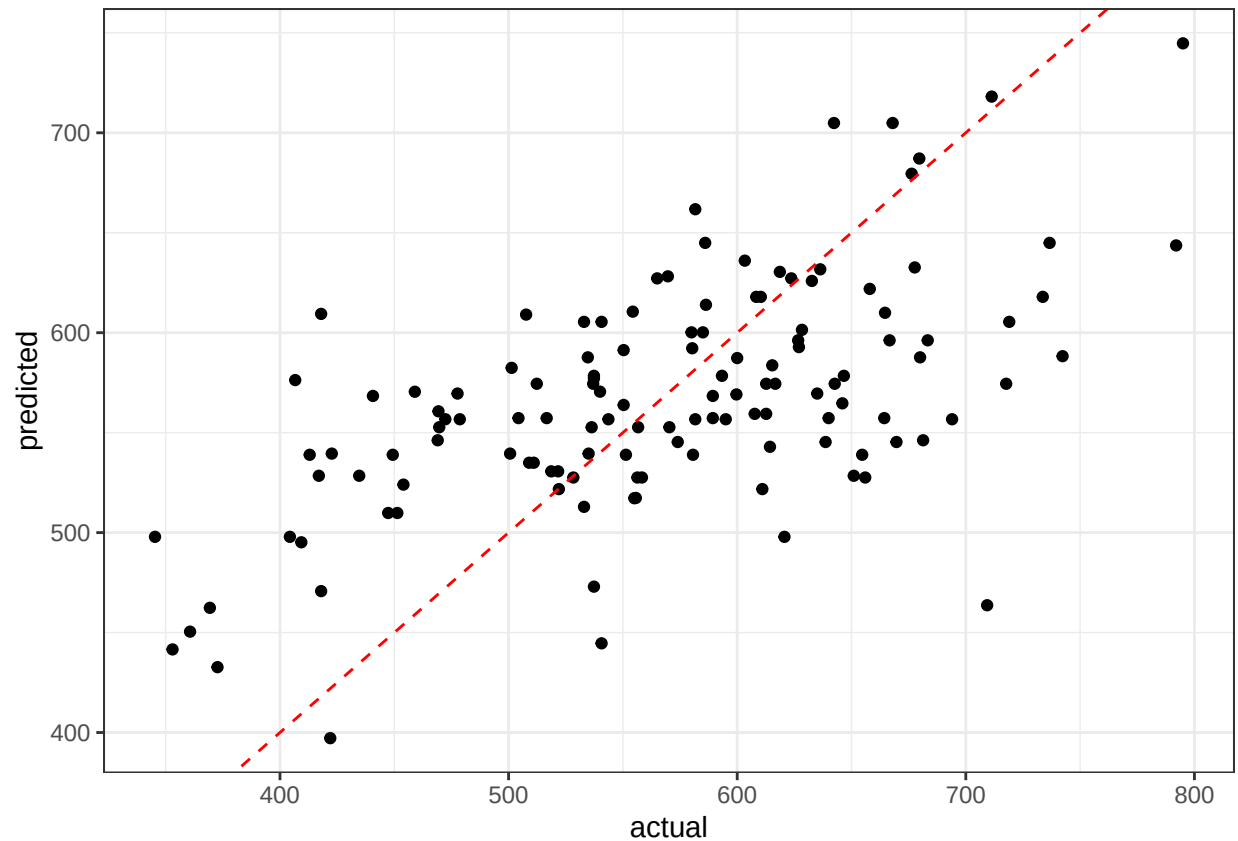
```



```
## BIC model
rmse(predict(BICmod3_weighted), Survey_cleaned$avg_time)
```

```
## [1] 76.78642
```

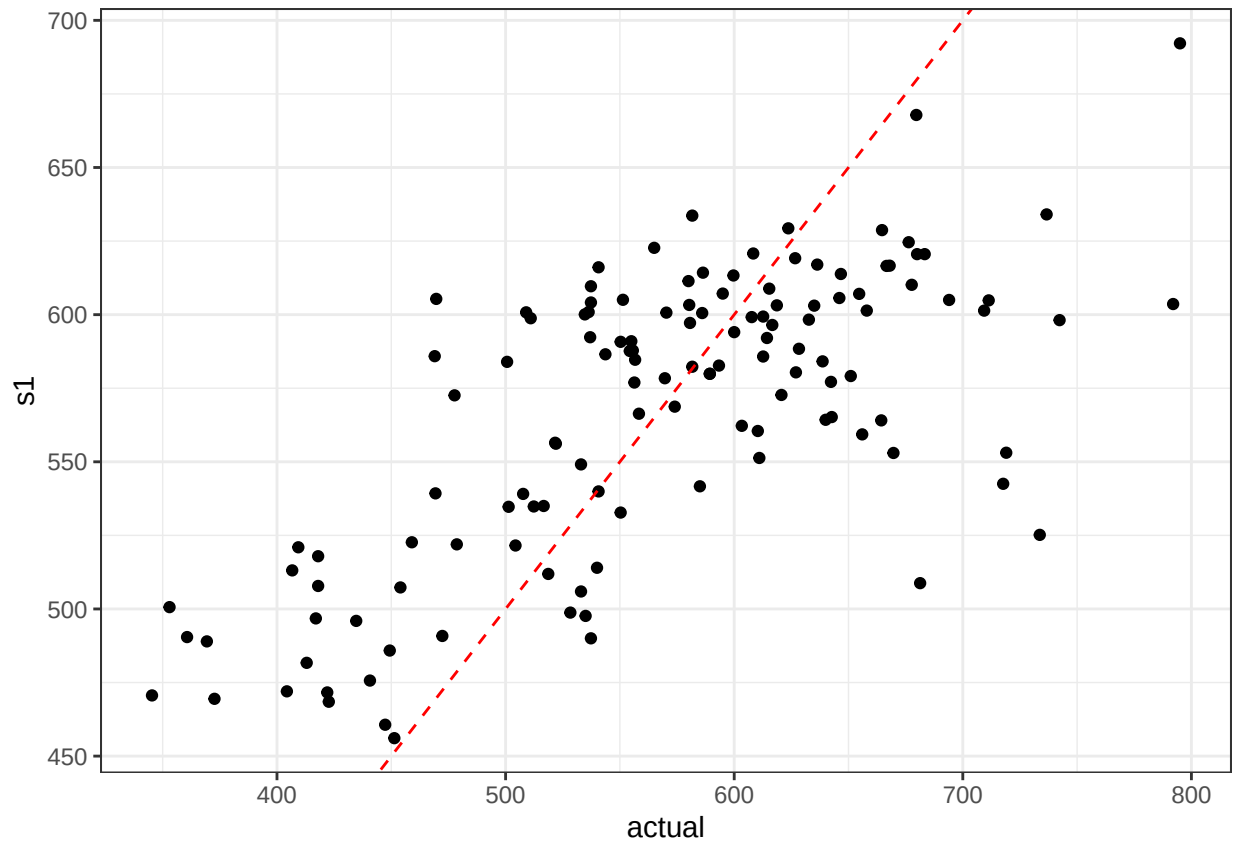
```
fitted_BIC <- data.frame(
  actual = Survey_cleaned$avg_time,
  predicted = predict(BICmod3)
)
fitted_BIC %>% ggplot(aes(x = actual, y = predicted)) +
  geom_point() +
  geom_abline(intercept = 0, slope = 1, color = "red", linetype = "dashed") +
  theme_bw()
```



```
## Lasso model  
rmse(predlasso, Survey_cleaned$avg_time)
```

```
## [1] 70.01466
```

```
fitted_Lasso <- data.frame(  
  actual = Survey_cleaned$avg_time,  
  predicted = predlasso  
)  
fitted_Lasso %>% ggplot(aes(x = actual, y = s1)) +  
  geom_point() +  
  geom_abline(intercept = 0, slope = 1, color = "red", linetype = "dashed") +  
  theme_bw()
```



```
# Ridge model  
rmse(predridge, Survey_cleaned$avg_time)
```

```
## [1] 62.5103
```

```
fitted_Ridge <- data.frame(  
  actual = Survey_cleaned$avg_time,  
  predicted = predridge  
)  
fitted_Ridge %>% ggplot(aes(x = actual, y = s1)) +  
  geom_point() +  
  geom_abline(intercept = 0, slope = 1, color = "red", linetype = "dashed") +  
  theme_bw()
```